

Do time series foundation models know their tails? Output-form dependence in zero-shot Value-at-Risk and Expected Shortfall

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Abstract

We evaluate zero-shot Value-at-Risk and Expected Shortfall forecasts from five time-series foundation models on 32 daily return series. The core extraction and testing protocol was fixed in advance; pretraining contamination is kept separate. A tail forecast depends on how it is read off the output interface; the extraction rule is part of the experiment. Native deep-tail forecasts range from unusable by construction to partially calibrated; some rankings depend on the extraction rule. Contrasts within and across families link failures to output form; the evidence is associative, not causal. Regime failures are head-specific. Replacing the conditional scale helps all five models, most strongly for the weakest; location is second-order at the daily one-step horizon. A hybrid substituting the GARCH scale, keeping the model's location, meets the joint criterion on all five models and matches classical GARCH–EVT without exceeding it. Deep-tail model comparisons are not well defined without the extraction rule.

Keywords: Volatility forecasting, Calibration, Evaluating forecasts, GARCH models, Backtesting, Extreme value theory

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1. Introduction

The 1% return quantile that a zero-shot time series foundation model (TSFM) reports for tomorrow depends on two things. One is what the model has learned about the conditional distribution; the other is what its output head can express. Standard central-quantile evaluations largely conflate the two: heads that fit the center equally well can imply different tail quantiles. [Perez-Diaz et al. \(2025\)](#) argue that forecast form limits the operational questions a forecast can answer; we examine the consequences for financial tail risk. Our claim is that the probabilistic representation a head emits constrains whether a zero-shot tail forecast is identifiable. It also governs how sensitive the forecast is to the extraction rule and whether it can be repaired without training.

The evaluation covers one-step Value-at-Risk (VaR) and Expected Shortfall (ES) at $\alpha \in \{1\%, 2.5\%, 5\%\}$ for five models: Chronos-Bolt, Chronos-2, TimesFM-2.5, Moirai-2.0, and Lag-Llama. Forecasts are evaluated on 32 daily financial series from six asset classes and compared with seven econometric risk models. The protocol was fixed in advance: the extraction rules are pre-specified and the tail tests are calibrated by simulation. Evidence overlapping the models’ disclosed pretraining corpora is kept separate from strictly post-cutoff and synthetic evidence.

The paper reports three findings, together with the qualifications that condition them. First, the native deep tails either are structurally unusable or depend on the extraction rule. Chronos-Bolt clamps any request below its trained grid to the grid edge, which yields a 12.7% violation rate at a nominal 1%. Lag-Llama produces distinct tail quantiles, but none of its 96 asset–tail cells passes unconditional coverage under the GPD splice. The test runs at the 5% level, with the size calibrated by Monte Carlo. The ‘healthy’ quantile heads avoid those failures. Even so, their Student- t and GPD-splice rankings have Kendall $\tau < 0.5$ on 4 of the 31 assets comparable at $\alpha = 1\%$. Extraction is therefore part of what we measure. Throughout the paper, ‘healthy’ is shorthand for the quantile heads whose coverage stays inside this paper’s reference band under the defensible extraction rules. The label applies only to those coverage tests and is not a general endorsement.

By output form we mean the probabilistic representation the deployed head emits; it is our operational definition of forecast form. Within the contrasts the design allows, the observed failures are associated with output form. The evidence comes from two within-family contrasts (Chronos-2’s

quantile head versus Chronos-Bolt’s bounded grid versus Chronos-base’s token-sampling head, and Moirai-1.1’s mixture head versus Moirai-2.0’s quantile head) and one cross-form contrast (Moirai-2.0’s quantile head versus Lag-Llama’s parametric Student- t head). The Chronos lineage has three heads that share vendor and data lineage: a bounded grid, a token-sampling head, and a calibrated quantile head. Under the vendor’s default decoding settings, which we pin, the token-sampling head’s predictive scale collapses across a 12-asset sub-panel; on probed windows the shortfall is an order of magnitude. The Moirai pair is a within-family contrast between the earlier mixture and the later quantile output. The mechanism claim is limited to these contrasts; family and training corpus vary across the panel.

Volatility episodes add a qualification rather than a fourth finding. They do not produce a common TSFM response, and the better-calibrated quantile heads return to the calm-day band in roughly the same range of days as the econometric models. The under-dispersed parametric head is the slowest and the most heavily censored within the 60-day event window; this partially rejects the view, stated in advance, that regime fragility is a class property.

The second finding, from the repair experiment, indicates that the conditional scale is the binding shortfall. Conditional location is second-order at the daily one-step horizon. The scale point estimate is positive for all five models; the separation from zero holds under every clustered inference procedure only for the most under-dispersed head (Section 10). A hybrid arm, H, is a component-substitution diagnostic: it keeps the TSFM conditional location but uses the GARCH conditional scale before fitting an EVT tail. Training-free means that repairs estimate auxiliary components on the target series; no foundation-model weights are updated. H is the only training-free arm that satisfies the joint criterion (internally time-stamped; fixed after comparator values were visible but before any repair outcome) for 5 of 5 models. The third finding concerns the classical comparator: H matches GARCH–EVT and does not exceed it.

The paper makes three contributions.

- C1. *Interface and extraction identification*: a mechanism analysis based on the two within-family contrasts and the cross-form contrast, with the extraction rules E0–E3 as an experimental axis and a pre-committed test of the H2 token-quantization hypothesis.
- C2. *Tail-risk validation and regime evidence*: a comparison of TSFMs with econometric risk models using coverage tests, dynamic-quantile diagnos-

tics, and Model Confidence Sets; an ES layer reporting FZ0 loss and ESR backtests alongside effective tail counts $n\alpha$ and precision-fragile flags (Pele and Mazurencu-Marinescu-Pele, 2026a); and event-time calibration decay and recovery under a frozen episode rule.

- C3. *Component substitution and deployment implications:* the training-free repair arms F1–F4 and the hybrid H compared with GARCH–EVT under one protocol (Bayer and Dimitriadis, 2022; McNeil and Frey, 2000; Pele and Mazurencu-Marinescu-Pele, 2026a). A decision guide links each output form to its defensible extraction rule, to the minimum repair the evidence supports, or to a recommendation to replace the head.

Section 2 reviews related work. Sections 3 and 4 define the heads, data, and contamination controls, and Section 5 describes the evaluation design. Sections 6–9 present the empirical evidence, and Section 10 studies repair and the location–scale decomposition. Practical implications, limitations, and the conclusion follow.

2. Related work

Tail-risk evaluation and finite-sample ES. Goel et al. (2024) provide the closest empirical predecessor. They evaluate TimesFM for one-day VaR on an S&P 100 system, with the pretrained model assessed at its native decile. In their results, zero-shot use is significantly worse than the best econometric approaches. Fine-tuned variants improve violation-rate calibration and are comparable to GAS under quantile loss. Their study does not address ES, cross-head variation, mechanisms, or training-free tail repair. In our zero-shot scope we ask a different question: what can be learned about a head’s native or constructed deep tail when fine-tuning is not part of the workflow.

Lu (2025) surveys TSFMs in finance and proposes a framework that combines proper scores, VaR coverage tests, ES backtests, data-snooping controls, and regime stressors. Its only execution is an illustrative AR(1) demonstration. No TSFM is evaluated. We implement and extend that framework. The main additions are Monte-Carlo calibration for the deep-tail tests, whose practical effect is concentrated in the conditional-coverage layer (Section 5); an explicit comparison of the E0–E3 extraction rules; and finite-sample information reporting for ES.

Pele and Mazurencu-Marinescu-Pele (2026a) convert the $(n\alpha)^{-1/2}$ information rate into an operational precision audit. Their application covers 24

assets and four models, with no model-ranking objective. Their object is the comparison methodology; ours is the models. We adopt the effective-tail diagnostics and include their additive FZ0 recalibration as F4. Their application assigns each model its own extraction: Chronos-Small uses empirical sample summaries, TimesFM uses Student- t quantile matching, and Moirai uses a Student- t fit to predictive samples. Our E0–E3 sweep treats that extraction decision as part of the object being evaluated. They retain Chronos-Small as a stress case for severe first-stage VaR miscalibration. On our mechanism sub-panel we find that the same family’s token-sampling head shows a form of that failure mode that no extraction rule repairs under the vendor’s default decoding settings, which we pin (Section 9).

Forecast form and probabilistic calibration. [Perez-Diaz et al. \(2025\)](#) argue on design grounds that forecast form constrains downstream use and that a finite quantile grid does not identify behavior outside its trained levels. Our taxonomy refines that argument for tail risk by separating bounded grids, token-sampling heads, deep-quantile heads, and parametric heads. [Adler et al. \(2026\)](#) compare the calibration of recent zero-shot TSFMs on a fixed probability set. They evaluate central quantiles and leave deeper quantiles and distribution shift for future work. We add financial VaR, ES, and regime conditioning.

Conformal and post-hoc recalibration. [Achour et al. \(2025\)](#) combine zero-shot TSFM forecasts with split conformal calibration for symmetric prediction intervals on ERCOT and general forecasting benchmarks. [Aich et al. \(2025\)](#) study a different setting: return intervals and lower-tail VaR from gradient-boosted quantile forecasts, with a Robbins–Monro threshold update. Their base model is not a TSFM, and they do not evaluate ES. [Gibbs and Candès \(2021\)](#) provide the adaptive-conformal antecedent for this second approach. Our F2 arm tests whether online adaptation can correct native TSFM tail forecasts under financial regime change.

Two recent studies recalibrate deployed TSFM heads after the fact. [Pele et al. \(2026a\)](#) audit six zero-shot TSFMs and four econometric benchmarks on 24 assets at the 1% tail. They add a one-parameter conformal shift to the predicted lower quantile and use its magnitude as a signed audit statistic. Models separate into a signal-preserving regime, where the shift adjusts an informative base, and a replacement regime, where the correction is effectively the forecast. Their within-family Moirai contrast attributes the gap to the predictive interface, not the architecture.

The two ways of extracting tails from the same models reach opposite within-family verdicts. The two studies do not match on model revision, sign and scale conventions, or evaluation window, among other implementation choices. We report the size of the disagreement as an unresolved discrepancy and do not adjudicate it. The full reconciliation detail, including the raw numbers on both sides and our reproduction under their completion protocol, is in [Appendix B](#). None of this establishes an error on either side. It does establish that the choice of extraction layer can change the verdict. The extraction rule is an experimental axis in our design (E0–E3).

Their conformal construction targets VaR only, with ES as a supplementary diagnostic at 2.5%. The joint criterion here scores VaR and ES together. We ask instead which forecast component must be replaced when a scalar correction cannot restore coverage: location, scale, or tail.

[Cho and Matteson \(2026\)](#) document regime-dependent undercoverage of nominal 80% TSFM prediction intervals on four nonfinancial series. Their correction is multiplicative: a GARCH(1,1) model of the standardized forecast residuals re-scales the model’s own implied volatility, and in their comparison this outperforms adaptive conformal tracking. This is the closest methodological precursor to the hybrid H studied in [Section 10](#). Cho and Matteson retain the TSFM scale and correct central coverage multiplicatively. H instead discards the TSFM conditional scale, uses a GARCH scale with an EVT tail, and is evaluated jointly on deep-tail VaR and ES.

Financial applications, contamination, and uncertainty. A separate literature evaluates conditional means, point targets, and portfolio signals. Rahimikia, Ni, and Wang study excess-return prediction, directional accuracy, and long–short portfolios; off-the-shelf models generally underperform conventional benchmarks, with pronounced heterogeneity across models ([Rahimikia et al., 2025](#)). Point-forecast evaluations also include linear and log-return forecasts ([Noguer I Alonso and Franklin, 2026](#)), multivariate equity and interest-rate forecasts ([Das et al., 2026](#)), and transfer diagnostics for yield changes, realized FX volatility, and equity spreads ([Marconi, 2025](#)). These targets differ from joint VaR–ES validation. The results neither establish nor rule out deep-tail calibration.

[Li et al. \(2026\)](#) introduce dataset-level contamination auditing for TSFMs. Their probe detects exposure through adaptation dynamics, not documentary overlap; our disclosed-overlap comparison and post-cutoff strata are separate, documentary design choices. [Chinta et al. \(2026\)](#) propose a deep-evidential

output head that decomposes epistemic and aleatoric uncertainty. Their controlled empirical head comparison uses a common LSTM backbone, which makes it conceptually complementary to our evaluation of deployed heads.

Adjacent studies: prompt-based, point-forecast, and task-trained tail models. [Pele et al. \(2026b\)](#) prompt general-purpose language models for next-day return distributions and extract VaR and ES from the resulting samples. GPT-3.5, their best-calibrated model, approaches zero exceedances as the window grows; newer variants over-violate. The object is prompt-based numeric forecasting rather than the deployed time-series heads and extraction rules studied here. [He et al. \(2025\)](#) insert Chronos point forecasts into a historical-simulation window for a single crude-oil series. The tail estimate therefore remains a historical-simulation quantile, and no head or extraction comparison is involved. [Zhang et al. \(2026\)](#) train a purpose-built tail-risk model by FZ-loss minimization on the target equity panel. Its cross-market transfer is the only zero-shot element. That makes it a task-trained complement to our evaluation of pretrained general-purpose heads. [Brini \(2026\)](#) benchmarks zero-shot TSFMs against econometric specifications centered on the HAR family for realized-volatility point forecasts. Their Mincer–Zarnowitz recalibration attributes the short-horizon TSFM differences largely to level-and-scale calibration rather than volatility dynamics. Their finding concerns the conditional mean of volatility, and ours concerns the return tail, but both indicate the same component, the scale.

Methodological antecedents. F1 and H adapt filtered-residual EVT ([McNeil and Frey, 2000](#)). F1 uses the TSFM location and scale, while H combines TSFM location with GARCH scale. F4 is the additive joint-loss recalibration linked to [Bayer and Dimitriadis \(2022\)](#); [Pele and Mazurencu-Marinescu-Pele \(2026a\)](#). The backtests follow [Kupiec \(1995\)](#); [Christoffersen \(1998\)](#); [Engle and Manganelli \(2004\)](#) for VaR and [Fissler and Ziegel \(2016\)](#); [Patton et al. \(2019\)](#); [Acerbi and Székely \(2014\)](#); [Bayer and Dimitriadis \(2022\)](#) for ES. Forecast comparisons use [Diebold and Mariano \(1995\)](#); [Hansen et al. \(2011\)](#).

Existing studies do not jointly compare output heads, extraction rules, joint VaR–ES calibration, and component substitution in zero-shot financial forecasts; this paper examines that combination, together with the question of which component must be replaced when calibration fails.

3. Forecast forms and model panel

3.1. Models and benchmarks

Table 1 lists the evaluated heads. The five core models are Chronos-Bolt and Chronos-2 (Ansari et al., 2024, 2025), TimesFM-2.5 (Das et al., 2024), Moirai-2.0 (Woo et al., 2024; Liu et al., 2025), and Lag-Llama (Rasul et al., 2023). Each runs zero-shot at a pinned HuggingFace revision recorded in the run manifest; none is fine-tuned on the evaluation panel.

The comparison set contains seven econometric models: GARCH(1,1)- t (Bollerslev, 1986), GJR-GARCH- t (Glosten et al., 1993), EWMA(0.94) (J.P. Morgan and Reuters, 1996), historical simulation with 250- and 500-day windows, filtered historical simulation (Barone-Adesi et al., 1999), and CAViaR-SAV (Engle and Manganelli, 2004). A planned GAS- t check was dropped because no available implementation could be validated against a dependable external fixture. GARCH-EVT (McNeil and Frey, 2000) is used only in the repair comparison and reference-tested separately in Appendix B.

3.2. What each head can express

The unit of analysis is the output head: the interface that turns a learned representation into a probabilistic forecast. Its form determines which tail quantities are native and which must be constructed.

Chronos-Bolt: bounded quantile grid. The trained grid spans the deciles $0.1, \dots, 0.9$. A direct probe of the official implementation shows the 1%, 2.5%, and 5% requests clamped to the decile edge, so the tail levels are not distinguishable at the interface.

Chronos-2 and Moirai-2.0: quantile heads. Both official interfaces return distinct, ordered values at 1%, 2.5%, and 5%. That makes the native tail distinguishable; it does not establish its calibration.

TimesFM-2.5: decile-restricted quantile head. The implementation rejects requests outside its 0.1–0.9 grid, so it has no native deep-tail forecast. Every result reported for it depends on an explicit extraction rule.

Lag-Llama: parametric head. The network emits Student- t parameters, so it has an analytic quantile at any probability. The tail is distinguishable by construction, but it can still be under-dispersed when the emitted scale is too small.

Table 1: The audited TSFM heads: pinned revision, output-head type, native deep-tail behavior on a direct probe of the official implementation at $\alpha \in \{1\%, 2.5\%, 5\%\}$, and the registered pretraining data cutoff (calendar upper bound unless the vendor discloses one). Head type and native behavior are interface facts frozen before any backtest was run. The five core heads (upper block) run on the full 32-asset panel; the two mechanism sub-panel heads (lower block) run on the 12-asset cross-category subset.

model	repository @ revision	head type	native deep tail	cutoff
Chronos-Bolt	amazon/chronos-bolt-base @ 5d9f166	bounded quantile grid	clamp-to-decile edge	2025-11-21 (upper bound)
Chronos-2	amazon/chronos-2 @ 29ec376	quantile	distinguishable	2025-10 (upper bound)
TimesFM-2.5	google/timesfm-2.5-200m-pytorch @ 1d95242	quantile	reject-non-decile	2023-11 (disclosed)
Moirai-2.0	Salesforce/moirai-2.0-R-small @ 30f43ff	quantile	distinguishable	2025-11 (upper bound)
Lag-Llama	time-series-foundation-models/Lag-Llama @ 72dcfc2	parametric Student t	analytic-parametric	2023-10 (upper bound)
Chronos-base	amazon/chronos-t5-base @ ad294ea	token quantized sampling	distinguishable-token-quantized	2025-11-21 (upper bound)
Moirai-1.1	Salesforce/moirai-1.1-R-small @ 0c24ab9	mixture	distinguishable	2025-01-21 (upper bound)

The mechanism sub-panel, fixed before any results were seen, adds two contrast heads from the same model families, run on a 12-asset subset with two assets per asset class.

Chronos-base: token-quantized sampling head. The original Chronos form: returns are tokenized onto a value grid and forecasts are sampled token by token. A direct probe returns distinct, monotone deep quantiles. Under the pinned decoding configuration ($\text{top-}k = 50$), the returned one-step categorical distribution is truncated to at most 50 support points. That leaves about 3 distinct tail centers below the 5% quantile.¹ The extraction, fixed before evaluation, uses 1,000-sample empirical quantiles.

Moirai-1.1: mixture head. Moirai-2.0’s predecessor emits a mixture distribution. The official interface exposes it through samples only, with no analytic quantile path, so the same 1,000-sample extraction applies. The pair with Moirai-2.0 is a within-family change from mixture to quantile output.

A separate benchmark of the sample-generation method reports 9 exposed quantiles for Chronos-Bolt and 21 for Chronos-2 (Simeone, 2026). That supports the grid distinction, but the access path differs from ours: samples there are approximated from the exposed Chronos grids, and Moirai-2 sampling is restricted to probabilities in $[0.05, 0.95]$. Our E0 probes call the native quantile interfaces directly. We therefore use that evidence only for the two Chronos implementations and characterize Moirai-2.0 from our own probe.

3.3. Three diagnostic patterns

The extraction-rule sweep gives each failure mode a distinct pattern. A spread is the difference in mean violation rates between E0 and E2 at $\alpha = 1\%$ across the 32 assets. Chronos-Bolt has an 11.4 percentage-point spread: extracting a Student- t tail from its central grid produces a different forecast from the clamped native value. Lag-Llama’s native 5% VaR is violated on roughly 9.8% of SPX days, yet its spread is only 1.7pp. Because the emitted scale is narrow, alternative extraction rules applied to the same output change the forecast little. Chronos-2 and Moirai-2.0 show neither pattern, with

¹A single registry-probe statistic; the H2-sweep per-window median is 2 (Section 9).

spreads of 0.0pp and 0.7pp. The mechanism sub-panel adds a third pattern. Chronos-base’s token-sampling head violates a nominal 1% VaR on 32.4% of days and stays uncovered under every extraction rule. That is a scale collapse, not a clamp: no rule repairs it (Section 9).

The evidence has three parts: the core panel, the mechanism sub-panel, and two within-family contrasts (Chronos-2’s quantile head versus Chronos-Bolt’s bounded grid versus Chronos-base’s token-sampling head, and Moirai-1.1’s mixture head versus Moirai-2.0’s quantile head) and one cross-form contrast (Moirai-2.0’s quantile head versus Lag-Llama’s parametric Student- t head). The contrasts are diagnostic; they do not hold backbone and training data fixed. The Chronos triple shares vendor and data lineage, which makes the within-family comparison more informative but not causal.

4. Data and contamination controls

4.1. Asset panel and rolling design

The evaluation panel contains 32 daily series, drawn from a raw collection of 36. The four outside the panel are two quarantined FX pairs and the VIX and MOVE indices, which are used only as regime-dating inputs. The panel comprises eight equity indices, eight individual stocks, and four series each for FX, cryptocurrencies, commodities, and Treasury constant-maturity yields. EUR/USD and USD/JPY are excluded from every result because quality control found bad ticks in their 2008 vendor history, eight years before the 2016 evaluation start. The quarantine follows from a data-quality rule frozen before any results were computed, not from a claim of in-window contamination. It remains in force pending cross-verification against FRED H.10, so FX-group results apply only to the four retained pairs. Returns are measured as $r_t = 100 \Delta \log p_t$; Treasury yields enter as daily first differences in basis points rather than log returns.

The out-of-sample period begins on 2016-01-01, the evaluation start fixed in the pre-specified analysis protocol; end dates vary by series, and SPX ends on 2026-07-02. Series histories begin around 2000, so every model has a full history before its first target. Each TSFM receives the most recent 512 observations, matching the context interface the adapters share; the econometric models use a conventional 1,000-observation rolling estimation window, re-estimated every 21 trading days. Both lengths were fixed in advance. An earlier evaluation start would not reduce contamination exposure,

because pre-2016 history is within every model’s documented pretraining window.

The two window lengths do not select the same evaluation days everywhere. On five late-start series (csi300, eth, xrp, btc, ltc), the TSFM day sets exceed the baseline day sets by 459 to 488 trading days. Cross-family comparisons on those assets therefore pool slightly different day sets. Appendix Table D.11 reports every TSFM cell rescored on the baseline-day intersection; 12 of 285 unconditional-coverage decisions reverse, 9 toward passing and 3 toward failing.

All forecasts are one step ahead. The seed and configuration hash that identify the run are recorded in Appendix A. A single reviewed rolling module supplies every target date and exposes each adapter only to a read-only context array. The module’s built-in audit re-derives every window on every run. A sentinel test runs a synthetic adapter that reads ahead through the same module and confirms that the target observation and everything after it are unreachable. The repair pipeline has its own leakage sentinels, including a step-size sentinel.

4.2. Contamination controls

Public financial histories may have appeared in pretraining data, so we do not treat zero-shot status as evidence that a forecast is out of sample (Li et al., 2026). The design instead relies on three checks: cutoff bounds recorded in advance, an audit of disclosed-corpus overlap, and a synthetic arm with no historical overlap; post-bound strata are reported where the bounds allow them. None of these certifies an unobserved training corpus, and we do not claim that undocumented pretraining exposure has been identified or excluded.

Layer 1: cutoff bounds recorded in advance. Each model’s pretraining cutoff, or a calendar upper bound on it, was recorded before the forecast grid was run; the five bounds are between 2023-10 and 2025-11 (Appendix Table D.12). Only the TimesFM-2.5 date is vendor-disclosed; the others are conservative calendar bounds. Most of the sample predates the bounds: of 87,013 evaluation asset-days, the share on or before each bound ranges from 73.91% for Lag-Llama to 94.24% for Moirai-2.0. The bounds also follow two different rules. Chronos-Bolt’s is its pinned-revision date, while Chronos-2’s and Moirai-2.0’s are report-month bounds; under a unified revision-date rule, Chronos-2’s post-bound sample would shrink from 5,662 to 641 asset-days. Results are therefore

stratified when a regime episode is strictly after a model’s bound. The 2025-04 VIX episode is post-cutoff for TimesFM-2.5 and Lag-Llama; it provides a consistency check rather than a stand-alone post-cutoff experiment.

Layer 2: disclosed-corpus overlap. The disclosed training corpora were matched at component level to the 34 return series, that is, the evaluation panel plus the two quarantined FX pairs. Among 152 GiftEvalPretrain components, the only financial overlap is a Monash daily Bitcoin series ending around 2021-07. That component exposes Chronos-2, Moirai-2.0, and TimesFM-2.5 to BTC; no Bitcoin component is disclosed for Chronos-Bolt or Lag-Llama. Chronos-Bolt’s and Lag-Llama’s disclosed corpora instead have an FX-class exposure: both list the `exchange_rate` benchmark, whose eight daily currency series overlap this panel’s FX asset class over 1990–2010, entirely before the out-of-sample period. The audit cannot observe undisclosed augmentation or synthetic mixtures.

BTC therefore allows the only within-panel comparison with documented exposure. Under E2 at $\alpha = 5\%$ over the full out-of-sample period, Chronos-2 (exposed) and Lag-Llama (unexposed) both have a 6.43% violation rate, and the equality at the reported precision is genuine. The remaining rates are not ordered by disclosure status either: Chronos-Bolt has 5.66%, TimesFM-2.5 5.35%, and Moirai-2.0 4.14%.

The split date is 2021-07-01, the end of the disclosed Bitcoin component. The estimator is the plain two-group, two-period contrast on the group-mean violation rate $\bar{h}$ under E2 at $\alpha = 5\%$,

$$\widehat{\text{DiD}} = (\bar{h}_{\text{post}}^{\text{exp}} - \bar{h}_{\text{pre}}^{\text{exp}}) - (\bar{h}_{\text{post}}^{\text{unexp}} - \bar{h}_{\text{pre}}^{\text{unexp}}),$$

exposed = {Chronos-2, Moirai-2.0, TimesFM-2.5}.

Before the break, the exposed-group rate is 5.56%; after it, the rate is 5.03%. The unexposed-group rate changes from 6.03% to 6.07%. Over 1,966 pre-break and 1,830 post-break observations, the resulting difference-in-differences is -0.57 percentage points. The negative sign is the opposite of the pre-specified contamination signature, under which the exposed models deteriorate relative to the unexposed models once the disclosed component ends.

The exercise has limits. The set of models is small and heterogeneous, and the group contrast is reported without sampling-based uncertainty. The pre-specified protocol registered three DiD metrics (the hit-rate deviation, the quantile score, and the FZ0 loss) and required a contamination signature

to be robust across at least two. This exercise reports the hit-rate metric only; that requirement remains unexecuted. By construction, the exposure grouping is also collinear with head quality: the three exposed models are exactly the three with non-pathological heads, so any group contrast mixes disclosure status with head quality (Section 12). The BTC split is a descriptive single-asset contrast and does not identify a causal contamination effect.²

Layer 3: synthetic oracle. The final check has no historical overlap: every model forecasts simulated GARCH- t and GJR- t paths for which the conditional VaR and ES are known analytically. This arm verifies the forecast-to-backtest pipeline and is used to choose between extraction rules when empirical rankings differ, but synthetic fidelity does not establish which extraction rule is best for real returns.

5. Evaluation design

5.1. From a forecast head to VaR and ES: rules E0–E3

For a lower-tail probability α , let $v_{t,\alpha}$ denote the VaR forecast and $e_{t,\alpha}$ the corresponding ES, with $e_{t,\alpha} < v_{t,\alpha}$. Because native outputs differ in kind across heads, we fixed four extraction rules before running the full panel:

E0 (native). The requested output from the official implementation, including any native extrapolation. Chronos-Bolt’s clamped boundary is its E0 forecast; TimesFM-2.5 has no E0 because its head rejects off-grid probabilities.

E1 (normal tail). Central quantiles set a normal scale, giving $q_\alpha = q_{50} + s_N z_\alpha$ with $s_N = (q_{hi} - q_{lo}) / (z_{hi} - z_{lo})$ over the scale anchors below. The rule is intentionally misspecified and is retained only as a diagnostic.

E2 (Student- t tail). The same central anchors set the scale under a Student- t distribution, $q_\alpha = q_{50} + s_t t_\alpha(\nu)$ with $s_t = (q_{hi} - q_{lo}) / (t_{hi}(\nu) - t_{lo}(\nu))$, and the corresponding ES uses the closed-form Student- t tail integral. Degrees of freedom are fitted by maximum likelihood on the rolling context every 21 windows and clipped below at $\nu = 2.1$.

²The split was pre-specified as a contamination check with a 2021-07 break; the reconstruction uses the E2 module unchanged and reproduces all five full-window rates to the reported precision before the sample is divided. The analysis conditions, fixed in advance, and the run identity are summarized in [Appendix A](#).

E3 (GPD splice). A peaks-over-threshold GPD is fitted below the threshold u , the context’s fixed $\tau_{\text{anchor}} = 0.10$ quantile (Pickands, 1975; Embrechts et al., 1997; McNeil and Frey, 2000). The fit population is the kept exceedances: context observations strictly below u , after removal of quantization ties at the threshold. The ties’ probability is counted as a point mass at the body–tail boundary and enters no deep-tail formula. The GPD’s unconditional weight is the empirical mass of its fit population, $\tau_{\text{mass}} = n_{\text{kept}}/W$, measured per rolling window ($W = 512$). With fitted GPD shape ξ and scale β , the context-tail quantile is

$$q_{\alpha}^c = u - \frac{\beta}{\xi} \left[\left(\frac{\tau_{\text{mass}}}{\alpha} \right)^{\xi} - 1 \right]$$

(with the exponential form as the $\xi \rightarrow 0$ limit). One definition applies to all 32 assets. On the 28 continuous series no threshold ties occur and the kept count is constant, $\tau_{\text{mass}} = 52/512$; on the four rate series quantization ties occur, the per-asset mean masses are 0.082–0.088, and per-window values straddle the anchor. The per-window masses are logged with the fits. The model-space forecast applies the location–scale map anchored at $(q_{50}, q_{0.10})$,

$$q_{\alpha} = q_{50} + a (q_{\alpha}^c - q_{50}^c), \quad a = \frac{q_{0.10} - q_{50}}{u - q_{50}^c},$$

where q_{50}^c is the context median. ES uses the GPD mean-excess expression below the α -quantile, with the same τ_{mass} on the same path, mapped through the same (q_{50}, a) . When the fitted shape reaches $\xi \geq 1$, the implied ES does not exist and is left undefined. The empirical-mass convention corrects a normalization inconsistency identified during a revision round; pre-correction archives are retained.

The scale anchors are (0.25, 0.75) for four heads; TimesFM-2.5’s decile grid falls back to (0.2, 0.8).

The rules use different subsets of the head’s output, and each information set bounds the part of the model to which the corresponding result can be attributed. E2 depends on the head only through the central anchors $(q_{\text{lo}}, q_{\text{hi}}, q_{50})$. E3 uses only $(q_{50}, q_{0.10})$; its threshold, GPD parameters, and extrapolated quantile all come from the shared rolling context, so its output is algebraically invariant to the grid’s values below the decile anchor. E0 is

therefore the only rule that reads a head’s native deep tail. Under E1–E3 the scored forecasts are reconstructions built from the context and anchored at the head’s center.

Sampling heads use $S = 1,000$ draws. The pilot measures the resulting Monte-Carlo error: at the 1% quantile, mean absolute error falls from 0.233 to 0.046 return points as S increases from 250 to 4,000. The GPD splice has a second uncertainty source: the bootstrap standard deviation of an extrapolated 1% quantile at context length 512 is 0.338 return points. These quantities are the reference scale for extraction differences; they are not compared directly with rank correlations.

5.2. Backtests and forecast comparisons

VaR diagnostics comprise the Kupiec unconditional-coverage test (Kupiec, 1995), Christoffersen independence and conditional coverage (Christoffersen, 1998), and the dynamic-quantile test with four lags (Engle and Manganelli, 2004). ES is assessed with the zero-homogeneous FZ0 loss (Fissler and Ziegel, 2016; Patton et al., 2019),

$$\text{FZ0}_t = \frac{\mathbf{1}\{y_t \leq v_t\} (y_t - v_t)}{\alpha e_t} + \frac{v_t}{e_t} + \log(-e_t) - 1, \quad e_t < v_t, \quad e_t < 0,$$

whose degree-zero homogeneity means that rescaling returns and forecasts by a common factor shifts every forecast’s loss by the same additive constant, so loss differences are unchanged. The ES layer further includes the Z_2 statistic of Acerbi and Székely (2014), summarized in Appendix Table D.9, and the strict ESR regression backtest of Bayer and Dimitriadis (2022). ESR uses a copy of the authors’ reference implementation, checked into the repository.

Deep-tail samples make asymptotic reference distributions unreliable. For each realized sample length and α , we therefore simulate the null distribution of the coverage statistic from iid Bernoulli hits; results use 10,000 null replications and a 0.05 test level. The procedure has precedent in VaR backtesting: Christoffersen and Pelletier (2004) document that the asymptotic reference distributions of standard coverage tests are unreliable in realistic risk management sample sizes and compute exact Monte-Carlo p -values instead. At this panel’s sample lengths, the correction matters mainly in the conditional-coverage layer. The asymptotic and MC references disagree on 13 of the evaluation panel’s 2,496 unconditional-coverage decisions, all toward MC acceptance; at $\alpha = 1\%$ the MC reference changes 31 of 832 CC decisions, all to MC rejection. Condition (a) of the joint criterion, defined below, uses both

UC and CC decisions, so its outcome depends on the MC reference through the CC component. The DQ test uses its asymptotic reference distribution (Table D.6).

Loss comparisons use two-sided Diebold–Mariano tests with the one-step uniform-kernel variance and the small-sample correction; Model Confidence Sets are reported at the registered 10% and 25% levels (Diebold and Mariano, 1995; Hansen et al., 2011). The protocol therefore separates two roles that Nolde and Ziegel (2017) distinguish: calibration backtests of a single forecast stream, and comparative, score-based ranking across streams. The ranking role depends on the joint elicibility of the (VaR, ES) pair. Aggregation across assets reports descriptive fractions of cell-level or asset-level test decisions, as stated per table; the joint criterion of Section 10 is evaluated at asset level at $\alpha = 1\%$ only.

The core VaR and ES calculations were checked against external references before scientific use. Kupiec and Christoffersen match `rugarch`; ESR and exceedance tests match `esback`. FZ0 is locked to the published HMD_ES replication code (Pele and Mazurencu-Marinescu-Pele, 2026b) at commit 2490253e.³

For CAViaR-SAV, multi-start quantile regression runs in a transformed parameter space that enforces $|b_1| < 1$, so the SAV recursion is contractive by construction. Every candidate fit must also pass automatic path-validity screens before entering the start pool. On the evaluation panel there are zero failed fits and zero invalid forecast days across all 32 assets.

All rolling windows follow the observed row order of each asset. No interpolation creates synthetic trading days, and a missing target or context is skipped and left unfilled; forecast functions receive only the history preceding the target. Repair functions obtain past aligned forecast–realization pairs through the same restricted calibration view.

5.3. Finite-sample information in ES

The effective number of tail observations is $n\alpha$, not the raw sample length. Following Pele and Mazurencu-Marinescu-Pele (2026a), we report $n\alpha$ and the order- $(n\alpha)^{-1/2}$ precision scale for every ES cell. The main tables use $n\alpha < 10$ as a cell-level low-information flag. The flag is not a formal test of

³That repository’s `kupiec.p` helper and our implementation disagree on the null log likelihood. Ours agrees with `rugarch`. This difference is recorded in the fixture set and has no effect on FZ0.

an individual FZ0 gap, so close loss values are not given a fine ranking merely because both cells exceed that threshold.

At $\alpha = 1\%$, the full-window median is near 26; the shorter repair sample lowers the typical count to about 21. At this design’s sample lengths the flag is a reporting convention rather than an active screen. $n\alpha$ is determined by the sample length and α ; the panel minimum is 21.6 and the matched-sample minimum 16.6, so no cell can reach the floor of 10. We report the counts in the result tables and interpret the FZ0 evidence in coarse groups.

5.4. Repair sample and pre-specified joint criterion

The pre-specified repair arms are F1, F2, F3, F4, and H. F1 and H reconstruct the tail; the others transform an extracted forecast. All arms use a rolling history of $W = 500$ out-of-sample days and update every 21 days. All arms begin producing forecasts immediately, but the common evaluation sample begins on out-of-sample day 501. Within each model–asset pair, every arm and comparator is scored on the same dates; the median matched sample contains 2,139 observations.

The repair-side EVT tail keeps 0.10 as both threshold anchor and extrapolation mass. Across all of its $W = 500$ standardized-residual calibration windows the empirical threshold mass is exactly 0.10 with zero exceptions, so the fixed and empirical-mass conventions coincide there.

The pre-specified joint criterion is evaluated at $\alpha = 1\%$. Condition (a) bounds the asset-level MC UC+CC pass rate relative to FHS. Condition (b) bounds per-asset FZ0 relative to the better of GJR- t and FHS on a required fraction of assets. For an arm A with per-asset mean FZ0 $\bar{\ell}_i(A)$,

$$(a) \quad \frac{1}{32} \sum_{i=1}^{32} \mathbf{1}\{\text{UC}_i(A) \wedge \text{CC}_i(A)\} \geq \frac{1}{32} \sum_{i=1}^{32} \mathbf{1}\{\text{UC}_i(\text{FHS}) \wedge \text{CC}_i(\text{FHS})\} - 0.10,$$

$$(b) \quad \frac{1}{32} \# \left\{ i : \bar{\ell}_i(A) \leq 1.05 \times \min(\bar{\ell}_i(\text{GJR-}t), \bar{\ell}_i(\text{FHS})) \right\} \geq \frac{2}{3},$$

where UC_i and CC_i are the MC-calibrated 5%-level decisions on asset i . A missing repaired ES fails (b) for that asset, and the registered fallback applies the tolerance additively (+0.05) on the rare asset whose reference FZ0 is non-positive. Success is the conjunction of (a) and (b). The earlier unmatched sample is reported only as a sensitivity check. Throughout, *pre-specified* means fixed in the version-controlled analysis protocol before the

corresponding results were computed; this reflects internal version control, not third-party registration.

The full Stage-4 comparator table was first assembled on 2026-07-07, and the joint-criterion thresholds were set the same day, knowing Stage-4 comparator values but before any repair outcome. The hybrid arm H was added to the design as a same-day amendment at the Stage-5 approval. The mechanism sub-panel was authorized and executed once, on 2026-07-16. Its size and its two within-family contrasts were pre-specified in the proposal; the twelve asset identities were not. The decomposition control arms of Section 10 (the fourth 2×2 cell and the zero- and constant-location arms) are post-hoc robustness checks suggested by a reviewer during the revision round, not pre-specified arms.

5.5. Volatility episodes and recovery

The episode rule was frozen before event-time trajectories were examined; Section 8 discloses the full freeze chronology and bounds the leakage. An onset requires three consecutive closes. Each must exceed

$$\max\{1.5 \times \text{trailing-250-day median, floor}\}.$$

The floors are VIX 25, MOVE 100, and annualized crypto realized volatility 75%. Episodes last 60 trading days, and onsets within 20 days are merged. The frozen rule yields 28 episodes: 11 VIX, 3 MOVE, and 14 crypto.

Inference uses calm-day null bands. A day-cluster bootstrap with 2,000 replications draws from calm runs of at least 60 days, recenters the distribution at nominal α , and takes a one-sided 95% bound. The primary recovery time τ_{rec}^{21} is the first event day $t \in \{21, \dots, 60\}$ on which the trailing-21 pooled violation rate enters the band from above. A parallel τ_{rec}^{10} is retained as a window-sensitivity check. Censoring and the detectable multiple of nominal are reported with every median.

6. Value-at-Risk results

Throughout this section, *UC pass rate* means the share of the 96 asset–tail cells ($32 \text{ assets} \times \text{three } \alpha \text{ levels}$) at which the Kupiec unconditional-coverage test is *not* rejected at the Monte-Carlo-calibrated 5% level. The rate is a descriptive summary of cell-level decisions, not a jointly sized test. Each rate conditions on one extraction rule; rules are never averaged. Table 2 reports all head–rule combinations; Figure 1 shows mean violation rates at $\alpha = 1\%$.

Table 2: VaR audit: fraction of the 96 asset–tail cells ($32 \text{ assets} \times \alpha \in \{1\%, 2.5\%, 5\%\}$) at which the Monte-Carlo-calibrated Kupiec unconditional-coverage test does not reject, by forecaster and extraction rule. ‘param’ = econometric baselines (no extraction rule). TimesFM-2.5 has no E0 forecast because its official head rejects off-grid quantile levels. Higher indicates fewer rejections; the rate is a descriptive summary of cell-level decisions, not a jointly sized test.

forecaster	E0	E1	E2	E3	param
Chronos-Bolt	0%	0%	43%	20%	—
Chronos-2	78%	2%	59%	84%	—
TimesFM-2.5	—	5%	70%	70%	—
Moirai-2.0	50%	22%	62%	92%	—
Lag-Llama	0%	0%	5%	0%	—
GARCH(1,1)- t	—	—	—	—	66%
GJR-GARCH- t	—	—	—	—	68%
EWMA(0.94)	—	—	—	—	44%
HS-250	—	—	—	—	54%
HS-500	—	—	—	—	82%
FHS	—	—	—	—	100%
CAViaR-SAV	—	—	—	—	95%

The MC-calibrated test has limited power at these sample lengths. At the panel’s modal $n = 2,639$ and $\alpha = 1\%$, the 5%-level acceptance band spans 17 to 37 violations. Against a true violation rate of $1.25\times$ nominal, the exact-binomial power is 0.212. A high pass rate is therefore compatible with material per-asset dispersion. FHS passes every cell, yet its per-asset $\alpha = 1\%$ multiples span $0.76\times$ to $1.28\times$ around a mean of $1.07\times$, all inside the acceptance band; GJR-GARCH- t ’s mean of $1.27\times$ is near the band’s edge, consistent with its lower rate. The E3 means are $1.01\times$ for Chronos-2 and $0.97\times$ for Moirai-2.0. Mean calibration that close to nominal is as much as the evaluation can certify; separating it from cross-asset dispersion inside the band exceeds the per-cell test’s power.

6.1. Econometric models establish an attainable reference

Several econometric models set a reference band for the deep tail. FHS passes every cell and CAViaR-SAV 94.8%. HS-500, GJR-GARCH- t , and GARCH- t are between 65.6% and 82.3% (Table 2). Under the constrained estimation of Section 5, CAViaR-SAV passes at least 93.8% of cells at every

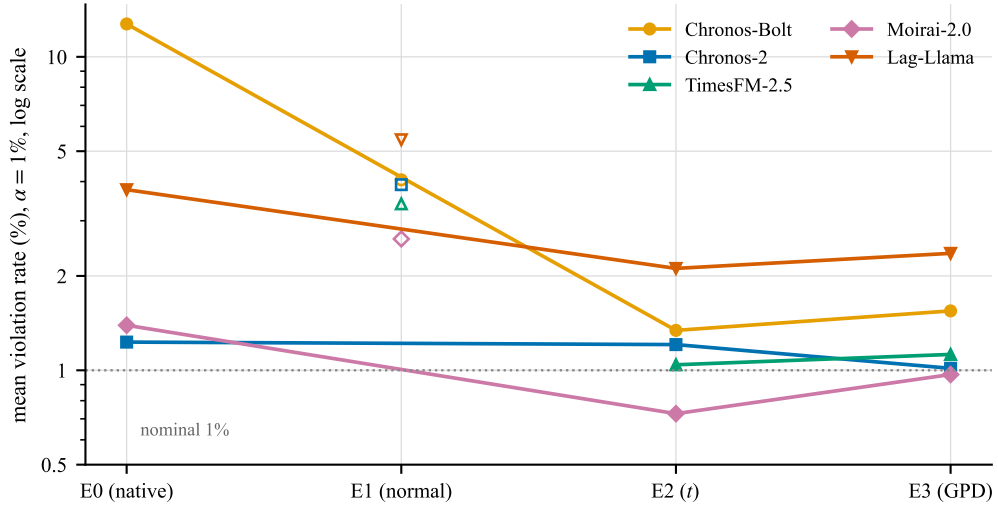


Figure 1: Mean violation rate at nominal $\alpha = 1\%$, on a logarithmic scale, by extraction rule and averaged over the 32 assets. Chronos-Bolt’s native value (E0) is 12.7%, the effect of its grid-edge clamp. Lag-Llama’s E0 value is 3.8%. Its tail is distinct but under-dispersed. The healthy quantile heads start near nominal and stay there under E2/E3. TimesFM-2.5 has no native E0 forecast. The dotted line marks the nominal rate. E1, the deliberately misspecified Gaussian control, is drawn as an isolated open marker and excluded from the connecting line, which runs E0 to E2 to E3.

tail level. HS-500 belongs to the band on the unconditional count only. With a CC pass rate of 29.2% and a DQ pass rate of 2.1%, its band position does not certify conditional adequacy. Acceptable deep-tail coverage is therefore attainable with the econometric models alone. The composition of the band matches the classical comparison literature: in the benchmark study of [Kuester et al. \(2006\)](#), a GARCH-filter-plus-EVT hybrid performs best overall, followed closely by filtered historical simulation.

One pair of econometric models isolates the tail form. EWMA(0.94) and FHS run the identical $\lambda = 0.94$ filter, seeded at the window sample variance.⁴ They differ only in the tail placed on that scale, a Gaussian quantile against the filtered empirical tail. That substitution moves the $\alpha = 1\%$ MC-UC pass count from 2 of 32 assets to 32 of 32; the mean FZ0 and the strict-ESR non-rejection rate move with it (Appendix Table D.7). Since the filter is shared, the difference within this pair is attributable to the tail form alone.

6.2. Results differ by head and extraction rule

In the common E3 column, Moirai-2.0 passes 92% of cells and Chronos-2 passes 84%. The gap is descriptive only. Under a paired cluster bootstrap over the 32 assets ($B = 10,000$, the three tail levels kept nested within each asset), the 95% interval for the difference, $[-18.8, +3.1]$ pp, includes zero. Within-asset correlation puts the effective number of independent units in the E3 columns at roughly 45 to 70.

TimesFM-2.5 passes 70% of cells under E3, and the interval for its E3 deficit against each better-calibrated head excludes zero: $[+4.2, +25.0]$ pp against Chronos-2 and $[+11.5, +33.3]$ pp against Moirai-2.0. Its E2 rate is the same 70%, and under that rule none of the pairwise differences among the three heads has an interval that excludes zero, so its ranking still depends on the extraction rule. Chronos-Bolt and Lag-Llama are much weaker under E3, at 20% and 0%. No single extraction rule can be labeled best for all heads. Moirai-2.0's E3 rate exceeds five of the seven econometric models and is below only FHS and CAViaR-SAV.

On the four rate series, the E3 test decisions are two-sided. Of the 20 rate-asset cells at $\alpha = 1\%$, 11 are not rejected, 5 are rejected for too few violations, and 4 for too many; all four over-hitting cells are Lag-Llama's.

All VaR rules are scored on identical finite-VaR day sets within each asset.

⁴The identity is verified by a unit test.

The ES-based statistics use separately reported validity sets, because ES can be undefined even when VaR remains valid; each ES table states its own denominator. An earlier implementation shortened, without warning, the E3 sample on the four rate series whenever the GPD ES was undefined, which dropped the heaviest-tail days and inflated the E3 pass rates. The correction restores the full 2,625-day sets on all four series and is disclosed in Section 7. Across families, each model is scored on its own delivered day set, and on the five late-start assets the TSFM sets are longer than the baseline sets (Section 4). Rescoring the TSFM cells on the baseline-day intersection flips 12 of 285 UC decisions in both directions (Appendix Table D.11), so the cross-family comparisons are not driven by the extra days.

6.3. Three failure modes: clamping, scale collapse, under-dispersion

The E0 columns show each failing head’s failure mode, and the modes differ. Chronos-Bolt violates its nominal 1% VaR on 12.7% of days on average, and every deep request returns the same trained 10% grid edge, so the output cannot encode tail depth. The spread between E0 and the Student- t rule E2 is 11.4 percentage points.

Lag-Llama has an analytic tail, but its nominal 1% forecast is violated on 3.8% of days, and the corresponding E0–E2 spread is only 1.7pp. Both rules use the scale the head emits. The two deep-quantile heads are closer to nominal: mean E0 violations are 1.2% for Chronos-2 and 1.4% for Moirai-2.0.

The mechanism sub-panel shows a third, more severe failure on its 12-asset subset. Chronos-base’s token-sampling head violates a nominal 1% VaR on 32.4% of days and, unlike the clamp, stays uncovered under every extraction rule. Section 9 discusses the three cases.

E1 is the misspecification check. Imposing a Gaussian tail reduces the Chronos-2 UC pass rate to 2%; the reporting rule, fixed before evaluation, excludes this control from headline rankings. For Chronos-Bolt, E1–E3 are extraction-based comparisons and E0 appears only as the clamping diagnostic.

Conditional diagnostics refine the unconditional result without reversing it. Under E3, the two deep-quantile heads pass the four-lag DQ test in 43% (Chronos-2) and 41% (Moirai-2.0) of cells, below the leading econometric models, whose DQ rates run from 46% (CAViaR-SAV) to 58% (GJR-GARCH- t). The MC-calibrated CC layer also does not reverse it: the same two heads’ 76% and 74% are at or above every econometric model except FHS and CAViaR-SAV (Appendix Table D.6). The deficit that remains against the leading econometric models is in violation clustering rather than in marginal

frequency. Because the DQ column uses an asymptotic reference that over-rejects at deep α (see the table note), the levels are informative and small differences are not.

7. Expected Shortfall results

7.1. FZ0 loss and regression backtests

Table 3 reports mean FZ0 at $\alpha = 1\%$, under E2 for the TSFMs and under the native specification for the econometric models. E2 keeps the common 32-asset panel. The E3 mean-excess ES is undefined on any day whose fitted GPD shape satisfies $\xi \geq 1$. For dgs30 such days are 2.3% of the sample, above the pre-specified 1% ES-validity margin. The E3 ES panel therefore aggregates on 31 of 32 assets, and the dgs30 cells are reported as missing. E3 ES evidence appears in the ESR results below and in the matched repair tables of Section 10.

The table also reports effective tail information: the median $n\alpha$ is 26.4, and no displayed cell is below the threshold of 10, fixed before evaluation. This supports broad separation, not fine ordering of nearby loss values.

The mean losses separate into groups. GJR-GARCH- t , GARCH- t , and FHS form the leading econometric group, with mean FZ0 between 1.799 and 1.814. The four quantile heads form a second group with higher mean loss than that group; Table 3 lists the per-model means. Lag-Llama is outside both groups at 2.112. The second group in turn has lower mean loss than the weaker econometric models, EWMA(0.94) and HS-500. These orderings are $\alpha = 1\%$ results. Within a group, the gaps are small relative to the tail-information scale.

The strict ESR test measures calibration; it does not rank losses. Under E3, Moirai-2.0 is not rejected at the 5% level in 85% of the asset-tail cells; Lag-Llama in 12%. Appendix Table D.8 reports the non-rejection grid across models and the rules E1–E3. E0 has no ES formula and no ESR cell. We use the authors’ reference implementation, included in the replication code, evaluated at its two-sided asymptotic p -value without bootstrap. The optimizer’s random restarts are seeded per cell, so reported p -values are exactly reproducible. The ESR column uses an asymptotic reference, whereas the UC and CC layers are calibrated by Monte Carlo because the design excludes their asymptotic references at deep-tail sample sizes. We therefore treat the column as a cross-check rather than a calibrated test.

Table 3: ES layer with finite-sample information diagnostics at $\alpha = 1\%$, E2 extraction (TSFMs) and native parametric forecasts (baselines): mean FZ0 loss over assets (lower is better), median effective tail count $n\alpha$, number of assets with a valid ES cell, and the number of precision-fragile cells ($n\alpha < 10$, the cell-level floor adapted from [Pele and Mazurencu-Marinescu-Pele \(2026a\)](#)). The $(n\alpha)^{-1/2}$ scale is an order-of-magnitude diagnostic, not a pairwise certificate; close FZ0 gaps carry no substantive ranking claim. ‘—’: CAViaR-SAV forecasts VaR only and has no ES cell.

forecaster	mean FZ0	median $n\alpha$	valid assets	fragile cells
Chronos-Bolt	1.890	26.4	32	0
Chronos-2	1.876	26.4	32	0
TimesFM-2.5	1.873	26.4	32	0
Moirai-2.0	1.884	26.4	32	0
Lag-Llama	2.112	26.4	32	0
GARCH(1,1)- t	1.800	26.4	32	0
GJR-GARCH- t	1.799	26.4	32	0
EWMA(0.94)	1.960	26.4	32	0
HS-250	1.988	26.4	31	0
HS-500	2.018	26.4	32	0
FHS	1.814	26.4	32	0
CAViaR-SAV	—	—	0	—

The FZ0-based Model Confidence Sets (Table D.4) discriminate weakly at these sample sizes rather than reproducing the loss split. EWMA(0.94) has the weakest ES results in the comparison, yet at the 10% level it remains in the set on every asset at all three α levels. The registered 25% level reduces its membership to 75% of assets at $\alpha = 1\%$ but leaves 94% at the shallower levels. The sets separate Lag-Llama: its membership falls with α at both registered levels, to 9% and 6% of assets at the shallowest level. HS-500 also declines, to 44% at $\alpha = 5\%$ at the 10% level. Membership also falls for the deep-quantile heads at the 25% level: Chronos-2 falls to 66% at $\alpha = 5\%$, below the leading econometric models there.

7.2. Extraction changes some deep-tail rankings

For each asset and α , the stability diagnostic, fixed before evaluation, compares the E2 and E3 model rankings by Kendall’s τ . The comparison covers the 93 cells where both rankings exist; dgs30’s three are excluded because its E3 FZ0 is invalid for all five models under the $\xi \geq 1$ ES-validity

rule disclosed above. The rankings have $\tau < 0.5$ in 8 cells. At $\alpha = 1\%$, the count is 4 of 31 and the mean correlation is 0.723. Half of the low- τ cells are at the deepest level, and all four are cryptocurrency or single-name equity series. The empirical conclusion is conditional on extraction: a defensible change in the extraction rule for a head can change the ordering.

The synthetic arm ([Appendix C](#)) provides a comparable oracle. On the simulated GARCH- t and GJR- t paths, E3 is closer to the analytic tail at $\alpha = 1\%$ for each of the four quantile heads it covers. Mean absolute ES errors are roughly 1.2–1.4 return points under E3, against 1.6–2.1 under E2. The arm arbitrates between E2 and E3 only; E0, E1, and Lag-Llama’s parametric head are not covered. Moirai-2.0 is the example cell of that comparison. This evidence is the basis for the deep-tail headline rule, but only under the synthetic DGP. E2 remains reported throughout because real returns have no such oracle.

7.3. When the GPD implies no finite ES

When $\xi \geq 1$, the E3 mean-excess ES is undefined and the cell is not imputed. Of the 586 $\xi \geq 1$ events recorded before the correction, 490 were parameter-boundary artifacts of maximum-likelihood GPD estimation. Tied exceedances from rate levels quantized at 1 bp push the estimator to that boundary. Under the tie-robust estimator the incidence falls to 85 windows, concentrated in the 30-year (61) and 2-year (23) series; none remain for the 5-year. We withdraw the earlier interpretation of this incidence as an empirical feature of post-2016 rate dynamics: it is predominantly a discreteness artifact. The remaining windows are genuine heavy-tail fits; their ES is undefined and the affected cells are reported as missing.

After conditional filtering, the boundary case is rare. It occurs once each in two asset-model cells (natgas–Chronos-Bolt under F1, SPX–Lag-Llama under H).

8. Calibration across volatility regimes

The pre-specified regime hypothesis, H3, predicted a class-wide pattern: TSFM calibration would deteriorate after a volatility transition and recover more slowly than parametric volatility models. The episode rule returns 28 events: 11 VIX, 3 MOVE, and 14 crypto realized-volatility episodes. Only the VIX pool supports the primary inference. At $\alpha = 5\%$, its trailing-21 null

band can detect a median violation rate of about $1.49\times$ nominal; the smaller MOVE and crypto pools are used for direction only.

The episode rule was frozen after full-period aggregate results were available but before any event-time trajectory was inspected. We therefore disclose and bound the leakage rather than treat this sequence as ex-ante registration. The thresholds follow common volatility conventions, the algorithm returns a closed event set without hand selection, and no trajectory used below had been examined when the rule was frozen.

8.1. Recovery times and censoring

Table 4 reports τ_{rec}^{21} , the first day $t \in \{21, \dots, 60\}$ at which the trailing-21 pooled violation rate returns to the calm-day band from above. TSFM forecasts use E3 in this section; econometric models use their native forecasts. A path still above the band at day 60 is censored; a path that never crosses the band is never-elevated and does not enter the recovery median. The band is estimated once, from all calm-day runs across the full out-of-sample period, including calm periods that postdate a given episode. Recovery classification is an ex-post descriptive convention, not an online monitor.

At $\alpha = 5\%$, under the trailing-21 definition, Moirai-2.0 and FHS recover fastest, with medians of 24.5 and 24.0 days.⁵ TimesFM-2.5 and Lag-Llama are the slowest, at 31. The remaining medians are between these values (Table 4). The ordering alone understates Lag-Llama’s weakness. Its recovered/censored/never count is 7/4/0: four of the eleven VIX episodes end before recovery is observed, so its median of 31 is a lower bound.

Episode-level uncertainty is wide. Bootstrap 95% intervals for the medians resample the recovered episodes with $B = 10,000$; they span roughly eight to twenty-three days per layer (Table 4). With eleven episodes, the ordering among the deep-quantile heads and the econometric references is uncertain.

The remaining censoring patterns are heterogeneous; Table 4 reports the counts. Two selection effects act on these counts, one favoring the well-calibrated models and the other favoring the weak model. Never-elevated episodes, most common for the well-calibrated models, truncate their exposure: an episode that never lifts a model’s rate above the band cannot show a slow recovery. Censoring truncates the observed outcomes of the weak model:

⁵Medians over an even number of recovered episodes fall on half-days and are reported at that native resolution. An earlier whole-day rounding gave 24 for both Moirai-2.0 and FHS, an exact parity that the raw values do not show.

Table 4: Event-time recovery after volatility-regime onsets: median τ_{rec}^{21} (trading days until the trailing-21-day pooled violation rate first re-enters the calm-stratum null band from above) over the eleven VIX-pool episodes, with an episode-bootstrap 95% interval for the median (recovered episodes resampled, $B = 10,000$), the interquartile range over the recovered episodes, and recovered / censored / never-elevated episode counts. An episode is censored when the rate is still above the band at the end of the 60-day event window; never-elevated episodes (rate never left the band) are excluded from the median. E3 extraction for TSFMs. Medians over an even number of recovered episodes land on half-days and are shown at that resolution; IQR and interval endpoints are rounded to whole days.

layer	$\alpha = 5\%$				$\alpha = 2.5\%$			
	med.	95% CI	IQR	r/c/n	med.	95% CI	IQR	r/c/n
Chronos-Bolt	30	[24,42]	24–40	11/0/0	30.5	[24,47]	24–44	10/1/0
Chronos-2	25.5	[23,34]	24–30	8/0/3	25	[23,35]	24–32	7/0/4
TimesFM-2.5	31	[24,42]	24–38	11/0/0	30	[24,42]	25–40	9/1/1
Moirai-2.0	24.5	[24,34]	24–31	8/1/2	29.5	[24,47]	25–45	8/0/3
Lag-Llama	31	[29,42]	30–37	7/4/0	29.5	[24,32]	25–32	8/3/0
GJR-GARCH- t	29	[23,45]	24–37	9/0/2	30	[23,43]	24–38	8/1/2
FHS	24	[22,38]	22–32	6/1/4	23	[22,35]	23–26	4/0/7

an episode still elevated at day 60 is excluded from the median. Cross-model medians therefore condition on different episode subsets, and in each case the subset removed is the one less favorable to detecting that model’s weakness. The recovery time is also doubly truncated by construction. It cannot be observed before event day 21, the trailing-window length, and it is right-censored at day 60, the episode span.

The deep-quantile heads are within the range of the two econometric references (Table 4). The under-dispersed parametric head is the clear adverse case, and the bounded and decile-restricted heads form an intermediate group rather than a uniform second failure group. This heterogeneity at $\alpha = 5\%$ leads to the H3 verdict: **partially falsified**. Regime fragility varies head by head rather than by model class.

Figure 2 shows the underlying paths for the COVID event and the 2025-04 event. For TimesFM-2.5 and Lag-Llama, the later episode is strictly post-cutoff and gives a directional consistency check on their relative behavior; it does not establish a post-cutoff ordering for the newer heads. During COVID, Lag-Llama’s trailing-21 rate reaches 0.395 at $t = 21$ against the nominal 5% level. The path shows both the initial undercoverage and the delayed recovery.

8.2. Volatility strata and Markov-state sensitivity

The same heterogeneity appears without event dating: the conditioning variable is trailing realized volatility measured through $t - 1$ (Figure 3). At $\alpha = 5\%$ under E3, Lag-Llama’s MC-UC pass rate falls from 69% in the lowest volatility quartile to 0% in the highest; all stratum figures are fractions of the 32 assets within each stratum. TimesFM-2.5 does not deteriorate in the lower strata but passes on only 47% in Q4. Chronos-2 is steady, at 88% in both the lowest quartile and Q4. Of the two econometric references, GJR-GARCH- t is at 88% in Q4 and FHS at 72%.

A two-state Markov classification gives a second conditioning scheme. Only lagged filtered state probabilities are used: smoothed probabilities would condition on contemporaneous information, and the predicted-marginal classification degenerates in this panel. The model converges for 28 of 32 assets, and all Markov-state pass fractions use that converged denominator. Chronos-2’s pass fraction is nearly the same in both states (93% low, 89%

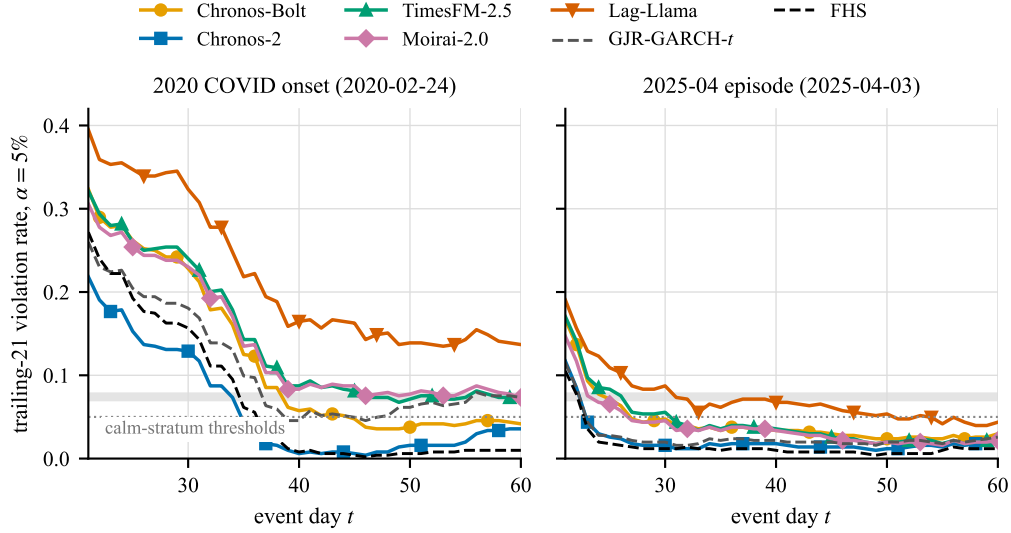


Figure 2: Trailing-21 violation rates at $\alpha = 5\%$ under E3 after the COVID VIX onset on 2020-02-24 and the 2025-04 VIX onset on 2025-04-03. The latter is strictly post-cutoff for TimesFM-2.5 and Lag-Llama. Dashed curves are GJR-GARCH- t and FHS. The gray strip spans one-sided 95% calm-day thresholds, computed per layer (0.070–0.078). Recovery classifications in Table 4 use each layer’s own threshold. At a nominal 5%, Lag-Llama reaches 0.395 at $t = 21$ in the COVID episode.

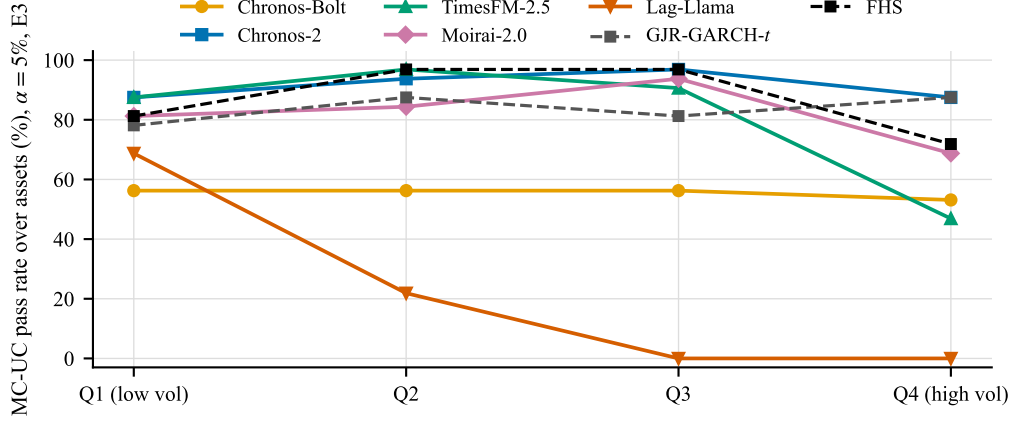


Figure 3: MC-UC pass rates at $\alpha = 5\%$, E3, by trailing-21 realized-volatility quartile. Conditioning is measurable at $t - 1$; econometric references are dashed. Lag-Llama deteriorates monotonically, TimesFM-2.5 weakens mainly in the top quartile, and Chronos-2 remains stable.

high). TimesFM-2.5 drops from 86% to 43% as the state switches. Lag-Llama passes on none of the assets in either state.

8.3. Regime-conditional performance of the repair

The F1 overlay gives a limited in-regime check. For Chronos-2 at $\alpha = 5\%$, trailing-21, the recovered/censored/never count is 3/0/7 among the usable VIX overlays. The missing episode is the earliest event, whose onset is too close to the out-of-sample boundary to construct the overlay; that leaves 7 of 10 usable paths never elevated under F1. The parallel τ_{rec}^{10} results and the smaller pools are retained in the replication material. When the two recovery windows disagree, we report window sensitivity without selecting the more favorable definition.

9. Evidence from the head contrasts

The results suggest a role for output form, but the strength of that statement depends on the comparison. A controlled head experiment would change only the output head and hold everything else fixed. The panel used here does not meet that standard. It has the two within-family contrasts and the cross-form contrast. The within-family pairs run on a common 12-asset

cross-category subset, and the evaluation-panel heads are re-sliced to the same subset, so no comparison mixes panel sizes.

Within the Chronos lineage. The strongest evidence in the lineage is the scale collapse of Chronos-base, the original token-sampling head. It violates its nominal 1% VaR on 32.4% of days on average across the 12 assets, and no extraction rule repairs it: unconditional-coverage passes are 0% under E0, E1, E2, and E3. Pass rates on the sub-panel are shares of its 36 asset-tail cells per extraction rule. Chronos-Bolt, the bounded grid, fails differently. Its E0 violation rate under the clamp is 12.6%, and the E0–E2 spread is 11.3 percentage points. Extraction partially repairs this head: under E2, 47% of its cells pass unconditional coverage. Chronos-2’s quantile head, the best-calibrated head in the lineage, shows a 1.3% E0 violation rate with a 0.0pp spread, and 83% of its E0 cells pass unconditional coverage.

The two failing heads differ in whether extraction repairs them. Clamping is an interface artifact, and extraction from the central grid can bypass it. The token-sampling collapse is not bypassed: the predictive distribution the pipeline delivers is too narrow, so changing the extraction rule does not correct it. Chronos-base’s E0–E2 spread of 9.6pp is similar to Chronos-Bolt’s, but it moves the violation rate from 32.4% only as far as 22.8%, and still no cell passes unconditional coverage.

Locating the scale collapse. A forensic analysis isolated the anomaly before the aggregate results were finalized. A standalone probe of the official pipeline, importing none of this paper’s harness code, reproduces the collapse under the same vendor defaults. On an SPX window the model’s 1% quantile is -0.093 where the realized context 1% quantile is -4.956 ; the gap on BTC and NVDA is of the same order. Three controlled probes test which inputs produce the collapse (Table 5). The model tracks a deterministic sine with tight uncertainty and a noisy trend with a correctly scaled spread. It collapses only on unpredictable iid noise.

A three-configuration decoding experiment on the mechanism sub-panel locates the failure (Table 6). It uses exact evaluations of the model’s one-step categorical CDF under the pinned vendor default $\text{top-}k = 50$, a relaxed $\text{top-}k = 500$, and no truncation. Under the pinned default the mean 1% violation rate is 32.34%, and no asset passes unconditional coverage at any tail level. With the truncation removed the mean falls to 1.41%, near nominal; per-asset dispersion remains, and the per-level pass counts are in Table 6. On the

Table 5: Chronos-base one-step forecasts on three controlled probes (official pipeline, vendor defaults, $S = 1,000$). A post-hoc forensic diagnostic, run after the calibration anomaly was flagged and before its measurements were accepted; provenance and the hypothesis tree are in the forensic memo. Spread is the predictive $q_{90} - q_{10}$ range.

probe	truth / realized	predicted median	spread
sine (period 20, amplitude 5)	-2.939	-2.931	0.023
trend (slope 0.05, noise 0.3)	$\sim +25.6$	+25.48	0.656
iid $t(5)$ noise (std ~ 1.24)	realized $q_{10} = -1.61$	-0.050	0.545

Table 6: Chronos-base by decoding configuration on the 12-asset mechanism sub-panel: mean violation rate of the nominal 1% VaR, MC-calibrated UC passes at the three tail levels, and the exact categorical-CDF 1% quantile on the SPX probe window (2021-03-31), whose realized context 1% quantile is -4.956 and whose stored Chronos-Bolt clamp value is -1.120 . One forward pass per window; the top- $k = 50$ row reproduces the delivered sampled E0 read (32.34% exact vs. 32.40% sampled).

decoding	mean hit ($\alpha = 1\%$)	UC 1%	UC 2.5%	UC 5%	spx q_{01}
top- $k = 50$ (pinned vendor default)	32.34%	0/12	0/12	0/12	-0.093
top- $k = 500$	7.59%	0/12	0/12	0/12	-1.488
untruncated	1.41%	6/12	10/12	11/12	-2.475

SPX probe window the exact 1% quantile moves toward the realized context quantile as the truncation is relaxed, and still does not reach it (Table 6). The exact evaluation under the pinned default reproduces the delivered sampled E0, which cross-validates our pipeline. The deep-tail collapse is therefore attributable primarily to the top- $k = 50$ truncation in the pinned default rather than to native model behavior. The reported results characterize the head as deployed. The untruncated distribution is near nominal on average and dispersed asset by asset.

The forensic probes showed that the collapse is not an artifact of our pipeline. The mass concentration they found on unpredictable series persists in the untruncated per-asset dispersion. The three-configuration experiment shows that the dominant term is the decoding truncation. The finding is consistent with the published stress case for this family. [Pele and Mazurencu-Marinescu-Pele \(2026a\)](#) retain Chronos-Small for severe first-stage VaR miscalibration. Our Chronos-base measurement is a severe first-stage miscalibration case in the same family and vintage, unrepaired here by any extraction rule under the pinned default.

Within the Moirai family. The second within-family contrast compares Moirai-1.1’s mixture head with Moirai-2.0’s quantile head. On these tests every difference favors the quantile head. The quantile successor is closer to nominal at the deep tail (1.38% against 1.86% mean E0 violation at 1%). Under E0, 42% of the successor’s cells pass unconditional coverage against 8% for the mixture. The E2 shares are 64% and 47%. The gap is widest under E3: unconditional passes are 89% against 8%, and conditional passes are 78% against 8%.

The loss-based results agree. Mean FZ0 is 1.955 versus 2.000, a difference within the coarse-group resolution of Section 7. The quantile score at the 5% level is 0.2855 versus 0.2993. Both favor the successor. The mixture’s shallower-tail behavior is associated with the gap. Moirai-1.1’s E0 violations grow from 1.9% at the 1% level to 7.5% at the 5% level; the tail is progressively too narrow. The mixture form’s theoretical advantage, a full predictive density, was not accessible through the official interface, which exposes samples only. In this family the change from mixture to quantile head is accompanied by better calibration on every dimension measured here.

Across forecast forms. Moirai-2.0 and Lag-Llama both return distinguishable tails, yet their E3 UC pass rates are 92% and 0%. These rates are shares of the 96 asset–tail cells, at the MC-calibrated 5% test level, pooled over the three tail levels. Lag-Llama’s analytic Student- t head is expressive enough for the deep-tail query, but its emitted scale is under-dispersed. A distinguishable tail is therefore not sufficient for calibration. Because model family and training data change at the same time, the contrast does not identify forecast form as the sole cause.

H2: token quantization and tail resolution. The pre-specified hypothesis H2 held that token quantization limits tail resolution. Its dedicated test, run under an analysis convention fixed before any result was seen, returns a verdict of not supported. Two of the three pre-committed criteria hold. The achievable deep tail is discrete: the median count of distinct token values below the 5% quantile is 2. The limit is form-specific. A fine-grained probability sweep resolves a median of 34 distinct deep-tail VaR values for Chronos-2, against 3 for Chronos-base and 1 for the clamped Chronos-Bolt.

The failed criterion identifies the source of the limit. The token grid itself is fine-grained, with a median nearest-neighbor gap of 0.015 return points, well inside sampling noise, so the resolution limit does not come from coarse

bins. It comes from the top- k truncation combined with probability-mass concentration on unpredictable series. There the delivered categorical keeps only two or three tail tokens. Under the pre-committed criteria the binding constraint is the decoding truncation rather than token quantization, and the discrete tail follows from it.

H1: bounded heads and extrapolation assumptions. The pre-specified hypothesis H1 held that bounded heads’ deep-tail output is a function of extrapolation assumptions rather than of model knowledge. It is supported jointly by the information-set analysis of Section 5 and by direct E0 evidence from the two restricted-output heads. The analysis shows that E0 is the only rule that uses a head’s native deep tail. The E0 evidence shows what the restricted heads return: a deep-tail request is clamped to the decile edge by Chronos-Bolt’s bounded grid and rejected as a non-decile level by TimesFM-2.5’s head. Chronos-base’s decoding-configuration dependence corroborates the broader interpretation that the delivered deep tail can be a property of the delivery layer. It is not direct H1 evidence, because the token-sampling head is not a bounded head.

Scope of the mapping. Within the available contrasts, three failure modes map to three output forms. The bounded grid clamps and cannot express the request. The token-sampling head can express it but, as decoded under the pinned default, collapses the predictive scale. The parametric head is analytically complete but under-dispersed. The two deep-deep-quantile heads are more stable across extraction. The same categories help describe regime behavior and reparability, but not as a one-to-one partition. Chronos-Bolt and TimesFM-2.5, for example, do not share Lag-Llama’s censoring pattern in the episode recoveries (Section 8).

The mapping remains associative. The Chronos heads share vendor and data lineage, which makes the within-family comparison more informative, but backbone, optimization, and training configuration still vary alongside the head. Claims in this paper are limited to the observed heads and the available contrasts.

10. Training-free repair

10.1. Repair arms

The comparison contains four repair arms and one decomposition arm. Each forecast uses only information exposed by the restricted calibration view:

a rolling history of $W = 500$ out-of-sample days, updated every 21 days.

F1: TSFM-filtered EVT The TSFM median provides location and its central-grid IQR provides scale; a POT-GPD tail is fitted to standardized residuals. This arm substitutes a new tail and never uses the head’s native deep-tail output.

F2: adaptive conformal An online additive offset updates the E3 VaR and ES after each observation (Gibbs and Candès, 2021). The step, fixed before evaluation, is $\gamma = 0.01\hat{s}$, with half-step and double-step sensitivity variants fixed alongside it. Here $\hat{s}$ is the head’s daily interquartile scale: the median of $(q_{75,t} - q_{25,t})/1.349$ over the first 500 out-of-sample days, computed once per series.

F3: multiplicative rescaling A single correction for each α is selected within the calibration window.

F4: additive joint-loss recalibration Separate VaR and ES shifts minimize in-window FZ0, following the additive recalibration used by Pele and Mazurencu-Marinescu-Pele (2026a) and related to Bayer and Dimi-triadis (2022). Inside the calibration window, a shift that violates the $e < v$ ordering receives infinite loss. Between refits the fixed shifts are applied to new days without that constraint, and where they violate the ordering out of window, the cell has no valid ES. That mechanism, not missing data, produces the excluded F4 cells in Table D.2.

H: scale-substitution hybrid H uses the TSFM median for location, the GARCH- t conditional σ_t for scale, and a POT-GPD residual tail; it shares the σ path with GARCH-EVT. Comparing F1 with H changes the scale source at fixed TSFM location; comparing H with GARCH-EVT changes the location source at fixed GARCH scale.

Classical GARCH-EVT is the external comparator; its filter and POT components are validated against independent fixtures (Appendix B). All results below begin at out-of-sample day 501, use the common matched dates, and have a median of 2,139 observations per asset; the evaluation is at $\alpha = 1\%$ under E3.

In this section, *coverage* means the fraction of the 32 assets passing both MC-calibrated UC and CC at the 5% level. The denominator differs from Section 6, where the pass rates pool the 96 asset-tail cells over three α levels.

10.2. The joint criterion and its chronology

The criterion has two conditions. Condition (a) requires coverage of at least 68.1%, the matched-sample FHS rate less 10pp. Condition (b) requires per-asset FZ0 no greater than $1.05 \times \min(\text{GJR-}t, \text{FHS})$ for at least 2/3 of all 32 assets.

H is the only arm that meets the joint criterion, and it does so on all five models; its per-model coverage rates and condition-(b) fractions are reported in Table 7. Every F1–F4 arm fails it for every model.

The joint result is threshold-sensitive. Over a grid of condition-(a) margins, FZ0 tolerances, and condition-(b) fractions, H passes on all five models in 14 of 36 combinations (Table D.16). The 5/5 verdict holds throughout the region at or looser than the pre-specified settings: margins of 10 pp or more with tolerances of 1.05 or more at all three fractions. Toward tighter thresholds the models fail in order: Chronos-2 first, below a 9.4 pp margin; Moirai-2.0, Chronos-Bolt, and TimesFM-2.5 below 6.25 pp. The model that fails first is also the best calibrated before repair, meaning the highest unrepaired matched coverage: Chronos-2’s 69%. Its hybrid H, also at 69%, has the smallest gap over the 68.1% threshold. At FZ0 tolerance 1.00 no model passes; the failure follows from the parity reported below.

F1 shows why coverage alone is insufficient. Its largest coverage gain, Lag-Llama from 6% to 69%, meets condition (a) but still fails the joint criterion. F1’s condition-(b) fraction is below two-thirds on every model (Table 7); replacing the tail while retaining the TSFM-IQR scale does not close enough per-asset FZ0 gaps.

The repair arms also start from unequal unrepaired coverage. Unrepaired E3 coverage is 69% for Chronos-2 and 6% for Lag-Llama (Table D.2). Chronos-Bolt’s unrepaired E3 and E2 coverages differ by 28pp, so its extraction dependence persists into the repair comparison. Figure 4 plots the matched-sample FZ0 means by head and arm.

10.3. Location–scale decomposition

Table 8 separates the F1 gap into a scale term, F1–H, and a location term, H–GARCH–EVT. The scale term is the larger of the two for every model: it ranges from +0.027 for Chronos-2 to +0.152 for Lag-Llama, while no location term exceeds +0.009.

The difference in magnitude supports the central interpretation: at the daily one-step horizon, location is a second-order term, and the point estimates attribute the larger part of the gap to the conditional scale; the strength of

Table 7: Pre-specified success criterion at $\alpha = 1\%$, E3, matched evaluation sample (out-of-sample day 501 onward, common day set): (a) MC-UC+CC pass rate over assets $\geq$ FHS $- 10$ pp, i.e. $\geq 68.1\%$; (b) per-asset FZ0 $\leq 1.05 \times \min(\text{GJR-}t, \text{FHS})$ on $\geq 2/3$ of all 32 assets (none fall below the precision floor at the matched n). Assets without a valid repaired ES cell (F4) count as failing condition (b); the denominator is always the 32 assets. The matched median n is 2139. The only arm that meets the joint criterion on any head is H, and it does so on all five. The final row applies the same two conditions to GARCH-EVT itself as a reference self-check. Table 8 decomposes the same matched-sample FZ0 gaps into scale and location terms. Rows here and in the other repair tables follow the repair-panel order (Chronos-2, Moirai-2.0, Lag-Llama, Chronos-Bolt, TimesFM-2.5); the audit tables of Sections 6–8 use the panel order of Table 1.

head	arm	pass rate	(a)	(b) frac	joint
Chronos-2	F1	78%	pass	0.62	fails
Chronos-2	F2	72%	pass	0.50	fails
Chronos-2	F3	53%	fail	0.28	fails
Chronos-2	F4	78%	pass	0.22	fails
Chronos-2	H	69%	pass	0.84	succeeds
Moirai-2.0	F1	62%	fail	0.47	fails
Moirai-2.0	F2	62%	fail	0.53	fails
Moirai-2.0	F3	50%	fail	0.19	fails
Moirai-2.0	F4	53%	fail	0.09	fails
Moirai-2.0	H	72%	pass	0.81	succeeds
Lag-Llama	F1	69%	pass	0.12	fails
Lag-Llama	F2	6%	fail	0.06	fails
Lag-Llama	F3	31%	fail	0.06	fails
Lag-Llama	F4	50%	fail	0.00	fails
Lag-Llama	H	81%	pass	0.78	succeeds
Chronos-Bolt	F1	69%	pass	0.53	fails
Chronos-Bolt	F2	38%	fail	0.38	fails
Chronos-Bolt	F3	50%	fail	0.25	fails
Chronos-Bolt	F4	59%	fail	0.09	fails
Chronos-Bolt	H	72%	pass	0.81	succeeds
TimesFM-2.5	F1	59%	fail	0.44	fails
TimesFM-2.5	F2	69%	pass	0.41	fails
TimesFM-2.5	F3	50%	fail	0.19	fails
TimesFM-2.5	F4	53%	fail	0.09	fails
TimesFM-2.5	H	72%	pass	0.78	succeeds
GARCH-EVT	(self-check)	72%	pass	0.88	(reference)

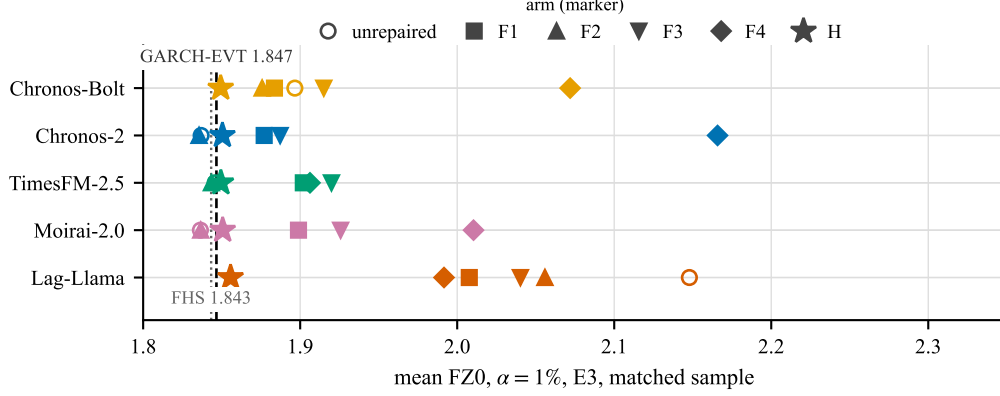


Figure 4: Mean FZ0 at $\alpha = 1\%$, E3, on the matched sample by head and repair arm. The dashed line is classical GARCH-EVT; the dotted line is FHS. H lies close to the GARCH-EVT line for each head. The additive-recalibration means use only assets with valid repaired ES cells, as reported in Table D.2. Marker shape denotes the repair arm and color the TSFM head; the open circle is the unrepaired forecast.

that inference differs across models. Across the five models, Across the five models, H’s mean FZ0 ranges from 1.849 to 1.856; the GARCH-EVT value is 1.847. H matches GARCH-EVT and does not exceed it. That is parity for a hybrid that uses a GARCH scale; it is not evidence that the native TSFM scale has been repaired. We claim no superiority for any repair arm over GARCH-EVT.

Appendix Table D.14 reports per-asset uncertainty for these means. The scale term exceeds the location term on a majority of the 32 assets for every model. The intervals come from an asset-class cluster bootstrap: the six asset classes are resampled with replacement as whole clusters, with a fixed seed and percentile bounds. The contrast is largest for Lag-Llama.

We repeat the inference under two further procedures: CR1 cluster-robust t with a $t(5)$ reference, and a restricted wild cluster bootstrap with full Rademacher enumeration over the six asset classes. Under all three procedures the scale term is positive for all five models. The number of models whose interval excludes zero ranges from one to four across procedures. Lag-Llama’s scale term, the largest of the five, is separated from zero under every procedure (restricted-WCR $p = .031$, the enumeration floor). Location terms are not separable from zero for four of the five models under any procedure. Lag-Llama’s location interval excludes zero under the primary cluster bootstrap

Table 8: Location–scale decomposition of the repair gap: mean FZ0 at $\alpha = 1\%$, E3, matched sample. F1 = TSFM location + TSFM-IQR scale; H = TSFM location + GARCH scale; GARCH–EVT = GARCH location + GARCH scale. The scale term (F1–H) isolates the scale source at fixed TSFM location; the location term (H–GARCH–EVT) isolates the location source at fixed GARCH scale. Differences are computed before rounding. Same sample, arms, and row order as Table 7.

head	F1	H	GARCH–EVT	scale (F1–H)	location (H–GE)
Chronos-2	1.877	1.850	1.847	+0.027	+0.004
Moirai-2.0	1.899	1.851	1.847	+0.048	+0.004
Lag-Llama	2.008	1.856	1.847	+0.152	+0.009
Chronos-Bolt	1.884	1.849	1.847	+0.034	+0.003
TimesFM-2.5	1.902	1.850	1.847	+0.052	+0.003

($[+0.001, +0.018]$) but not under the CR1 $t(5)$ or restricted-WCR procedures ($p = .125$); location remains second-order relative to scale.

A deterministic leave-one-class-out pass and a seeded unclustered bootstrap support this clustered inference. Every leave-one-asset-class-out mean keeps the scale term positive for all five models, and the unclustered asset-level bootstrap ($B=2,000$) excludes zero for all five scale terms. The clustered intervals are wider in eight of the ten model–term comparisons and are retained as the headline specification (Table D.13). The inference holds within the evaluated one-step decomposition.

Two post-hoc control arms, not pre-specified (Section 5), bound what the positive location term can mean. Replacing H’s TSFM location with zero, or with a constant unconditional location, yields mean FZ0 of 1.8457 and 1.8460, both marginally below GARCH–EVT’s 1.8467. A positive location term is therefore not evidence that the TSFM location is worse than the GARCH location: the decomposition cannot separate location sources at this magnitude. A fourth cell completing the 2×2 , GARCH location with TSFM scale, gives interaction terms between $-.003$ and $+.010$, negligible at panel scale.

10.4. Pairwise and set-based checks

The pre-specified double comparison pairs F1 with GARCH–EVT and with F4, using asset-level Diebold–Mariano tests on daily FZ0 differentials. At $\alpha = 1\%$, every F1–GARCH–EVT comparison that is significant at the 5% test level favors the comparator, and this holds under both variance

estimators. At $\alpha = 1\%$, the significant F1–F4 comparisons at the same level all favor F1. For Lag-Llama, under the Newey–West variance with automatic lag, the headline counts are 18/32 in favor of GARCH–EVT and 0/32 in favor of F1. The pre-specified baseline variance gives 19/32 and 0/32, retained in Appendix Table D.15.

The variance choice affects few decisions. The Newey–West variance changes 42 of the 960 asset-level decisions at the 5% test level, and 41 of them become insignificant. The two pooled-level exceptions that favor F1 are the same two under both estimators. Of the six pooled-level F1–F4 exceptions that favor F4 under the baseline variance, five remain significant under Newey–West, a strict subset of the baseline set (Table D.15).

Model Confidence Sets at the 10% level give the same broad result. H is a member for 97% of Lag-Llama assets and 100% for each of the other four models, while F1 membership for Lag-Llama is 47%. At the registered 25% level the set shrinks but retains the hybrid H: F3 and F4 memberships fall, and H remains a member in 97% of cells (Appendix Table D.5).

10.5. Sensitivity to the adaptation step

The F2 sensitivity analysis varies the adaptation step. For Lag-Llama, coverage under the $2\times$ step reaches 16%. Mean violation rates for the $0.5\times$, $1\times$, and $2\times$ steps are 0.0224, 0.0207, and 0.0186 at a nominal 1%. The corresponding mean FZ0 falls from 2.092 at the smallest step to 2.010 at the largest (Appendix Table D.3). Faster adaptation improves each measure but does not close the gap. The transformation arms can shift or rescale an extracted forecast; H instead replaces the conditional scale before the EVT tail is fitted.

The same limitation appears in two recent recalibration studies. [Pele et al. \(2026a\)](#) report that scalar conformal shifts improve marginal tail coverage without restoring conditional violation independence, and [Cho and Matteson \(2026\)](#) find [Gibbs and Candès \(2021\)](#)-style adaptive tracking conditionally unstable compared to an explicit volatility model.

10.6. Matched-sample sensitivity

Applying the same criterion to the earlier unmatched sample changes no F1–F4 outcome and one H outcome, Chronos-2+H. On the matched sample the arm exceeds the 68.1% threshold by +0.6pp; on the unmatched sample it misses the 80.6% threshold by –2.5pp. Condition (b) holds in both samples.

The change comes from the reference threshold. FHS coverage is 91% on the unmatched window and 78% on the matched window; the 500-day calibration burn-in removes the calm 2016–17 segment and changes the SPX sample from 2639 to 2139 observations.⁶

11. Implications for model choice

11.1. Implications of output form

Table 9 summarizes the practical guidance. A risk team should first ask whether the deployed head can express the requested tail, then choose a defensible extraction rule, and only then assess calibration.

11.2. Performance of the GARCH–EVT comparator

The comparator must itself satisfy the joint criterion, and on the matched sample GARCH–EVT does. Condition-(a) coverage is 72%, above the 68.1% threshold. The FZ0 condition (b) holds for 88% of assets, and the joint verdict is SUCCEEDS.

The headline uses the matched sample rather than a sample selected after the repair results were seen. On the earlier unmatched sample, GARCH–EVT coverage is 75% against an 80.6% condition-(a) threshold. Condition (b) remains 0.91. This is the same threshold effect seen for Chronos-2+H. FHS coverage, which sets the coverage threshold, changes from 91% on the unmatched sample to 78% on the matched sample for the burn-in reason given in Section 10.

11.3. Relation to previous empirical findings

Relative to Goel et al. (2024). Their study does not address ES, cross-head variation, mechanisms, or training-free tail repair. Our result is narrower than a claim that all zero-shot tails fail. The failure depends on the deployed head, and the alternative that succeeds is a hybrid that uses a GARCH scale.

Relative to Lu (2025). The present study applies that risk-aware evaluation framework. Two additions mattered in practice: Monte-Carlo critical values at deep tail levels, which helped mainly in the conditional-coverage layer, and explicit conditioning on the extraction rule.

⁶The matched sample is the pre-specified design; the unmatched sample is reported only as a sensitivity check.

Table 9: Selection guide from forecast form to usable risk product. The joint criterion is the internally time-stamped one of Section 5; H is a hybrid that uses a GARCH scale.

forecast form	evidence	empirical implication
Deep-quantile head (Chronos-2, Moirai-2.0)	Under E3, coverage is near the stronger econometric references, but rankings can change on fat-tailed assets at $\alpha = 1\%$ (§6–§7).	Report E2 and E3 separately. Use the E3 headline only with the synthetic-DGP qualification. Add H when the joint criterion is required (§10).
Decile-restricted head (TimesFM-2.5)	No native deep-tail query is available; all risk forecasts are constructed. The highest volatility stratum is weak (§8).	Treat extraction as part of the model specification. Use H for a criterion-grade hybrid, or switch heads when native tails are required.
Bounded, clamped grid (Chronos-Bolt)	The native nominal 1% forecast has a 12.7% violation rate, and deeper requests return the same boundary (§6).	Do not interpret E0 as a deep-tail forecast. H can form a criterion-grade hybrid; otherwise replace the head.
Under-dispersed Student- t head (Lag-Llama)	E3 UC pass is 0%, and calibration deteriorates with volatility. Recovery is slow and heavily censored (§8).	F2–F4 transformation arms do not close the gap, including the $2\times$ conformal step. F1 improves coverage but fails the FZ0 requirement. Use a hybrid that substitutes the scale, or switch heads (§10).
Any form, ES at $\alpha = 1\%$	Loss comparisons are limited by effective tail information (§7).	Report $n\alpha$ with FZ0 and avoid fine rankings below the available resolution.

Relative to [Pele and Mazurencu-Marinescu-Pele \(2026a\)](#). Their effective-tail framework transfers to a model evaluation. Our use of it is conservative: $n\alpha$ controls the resolution of the comparison and does not certify small loss gaps. Their application assigns one extraction procedure to each model; our extraction-rule sweep shows that the choice can matter. E2 and E3 change model rankings in 8 of 93 asset–tail cells. The additive joint-loss recalibration used as F4 does not meet our joint coverage-and-loss criterion. That finding concerns the harder joint target, not the average-loss objective for which the method was designed.

11.4. Contribution of the TSFM location component

At $h = 1$, holding the GARCH scale fixed, replacing the GARCH location with the TSFM median costs at most +0.009 FZ0 points in this experiment. That small location term is not evidence of broad location skill, and we claim nothing stronger. The zero- and constant-location variants, measured as post-hoc control arms, coincide with GARCH–EVT up to the negligible daily mean term (mean FZ0 1.8457 and 1.8460 against 1.8467, Section 10). The larger measured term comes from the conditional scale. H succeeds because the GARCH scale replaces the TSFM scale before the tail is fitted. The resulting product takes its location from the TSFM and its scale from GARCH; it is not a validated native TSFM tail.

Interpreting H as a scale substitution has a precursor in interval forecasting. [Cho and Matteson \(2026\)](#) correct TSFM prediction intervals by letting a GARCH(1,1) model of the residual process re-scale the model-implied volatility, and improve conditional 80% coverage on nonfinancial series. H extends the same diagnosis to the risk setting. Here the TSFM scale is replaced rather than re-scaled, and an EVT tail is fitted given the GARCH scale. Success is judged at deep-tail VaR and ES jointly under the internally time-stamped criterion. On that criterion, H matches classical GARCH–EVT and does not exceed it.

A volatility-forecasting counterpart supports the same interpretation. [Brini \(2026\)](#) finds that zero-shot TSFM differences in realized-volatility forecasting at short horizons are largely removable level-and-scale bias under Mincer–Zarnowitz recalibration. In their results, a genuine informational gain appears only for one small model at the monthly horizon. Their object is the volatility point forecast; ours is the distributional deep tail.

12. Limitations

1. *The horizon is one day.* All forecasts have $h = 1$. Aggregating to a 10-day horizon would introduce overlapping returns, horizon-dependent scale dynamics, and a different role for conditional location. The location–scale interpretation holds for the daily experiment only; no claim extends beyond $h = 1$.
2. *Extraction remains partly underdetermined.* The E0–E3 sweep makes the extraction choice explicit but does not settle it. Rankings under the two defensible rules differ in 8 of 93 cells. The E3 splice is also noisy: its bootstrap standard deviation at the 1% quantile is 0.338 return points (ctx=512). The synthetic arm favors E3 at that quantile, subject to the synthetic-oracle limitation below. Readers who do not accept that preference can use the E2 columns. The ES tables display effective tail counts and a cell-level low-information flag; they do not establish that every small pairwise FZ0 difference is identifiable.
3. *The recent post-cutoff sample is short, and the bounds mix two rules.* The cutoff bounds for Chronos-2, Moirai-2.0, and Chronos-Bolt are 2025-10/11, and the data end in 2026-07. Only for the two models with 2023 bounds is the 2025-04 episode strictly post-cutoff. The bounds are also set by two conventions: a pinned-revision date for Chronos-Bolt and report-based months for Chronos-2, Moirai-2.0, and Lag-Llama. Under a unified revision-date rule, Chronos-2’s post-bound sample would shrink from 5,662 to 641 asset-days (Appendix Table D.12). This layer is a consistency check; it does not support stand-alone post-cutoff inference. The disclosed-overlap audit cannot detect undocumented data. The BTC split is a descriptive contrast on a single asset and does not identify a causal contamination effect. Two further limits apply: the exercise reports only one metric, and the exposure groups are collinear with overall head quality. Both are stated with the exercise in Section 4.
4. *The oracle is synthetic.* On analytic GARCH- t and GJR- t paths the true tail is known, so we can observe how accurately each extraction rule recovers it. Real returns have no known true tail and may favor a different rule. Every conclusion that selects E3 over the Student- t rule has this qualification: the E3 preference is established on simulated processes and need not hold beyond them.
5. *The mechanism sub-panel is a 12-asset subset.* The pre-specified within-family extension ran on 12 assets spanning the asset categories, not

the full panel: Chronos-base and Moirai-1.1 have mechanism sub-panel evidence only, and their numbers are never pooled with 32-asset means. The available contrasts do not hold backbone, optimization, or training data fixed, so the attribution stays associative. For Chronos-base one more limit applies: the evaluated forecasts use the vendor-pinned sampling defaults. The three-configuration decoding experiment in Section 9 shows that this choice dominates behavior in the deep tail. For the token-sampling form, attribution to the head is therefore limited by the decoding configuration as well. H2 had its own pre-committed test and was not supported; the resolution limit that test found comes from the decoding truncation, not from coarse token bins (Section 9).

6. *The evaluation window starts in 2016.* The out-of-sample period begins on 2016-01-01. It therefore excludes the 2008–09 global financial crisis, and the regime evidence covers post-2016 episodes only.
7. *Context length is fixed at 512 observations.* The shared adapter interface fixes a 512-observation context in advance for every model. The econometric models estimate instead on 1,000-observation windows whose effective memory is far shorter than the nominal length. In the pilot ablation, Chronos-Bolt shows the same E0 clamp pattern at context lengths 256, 512, and 2,048. The effect of the window asymmetry is bounded by a post-hoc arm that re-estimates both GARCH variants and FHS on windows matching the TSFM context length: the GARCH pass rates decline moderately, FHS is essentially unchanged, and no econometric model leaves its performance class, so the reference band’s composition is not an artifact of the longer estimation history (Appendix Table D.17).
8. *The results have not yet been independently reproduced.* Several controls limit implementation discretion, among them external fixtures, pinned revisions, and a centralized rolling module; the replication package documents the rest. None of them substitutes for independent reproduction. The replication package is designed to make that test possible.

13. Conclusion

Across heads, native deep tails range from unusable by construction to partially calibrated, and some conclusions depend on how the tails are extracted. A bounded grid cannot distinguish requests below its trained

support, and a parametric head can be analytically complete yet under-dispersed. Even a healthy quantile head can change rank when a defensible extraction rule changes, so output form has to be checked before tail forecasts are compared. Only the native rule E0 uses a head’s own deep tail. Every other rule in this paper scores a reconstruction built from the context, and an evaluation must state which of the two it is scoring.

The second finding, from the decomposition, is that the point estimates uniformly implicate the conditional scale, and location is second-order at the daily one-step horizon. The scale term is positive for all five models; its separation from zero holds under every clustered inference procedure for the most under-dispersed head only. The largest location term is +0.009 FZ0 points, while the largest scale term is +0.152. These decomposition terms are specific to this experiment. They are not a general claim about the predictability of location.

Among the training-free arms, only the hybrid H satisfies the internally time-stamped joint criterion, and it does so on 5 of 5 models. The pass holds at the thresholds as set and at looser ones, and fails as they tighten; 14 of 36 grid combinations pass. H keeps the TSFM median but uses a GARCH conditional scale before the EVT tail is fitted. It matches classical GARCH–EVT and does not exceed it. The hybrid H passes; the unrepaired native head does not. Control arms that replace the TSFM location with zero or an unconditional constant perform comparably (Section 10). The pass is due to the GARCH scale and the EVT tail; the evidence does not establish a measurable location contribution at this horizon.

Two qualifications apply. Regime fragility is head-specific: the fastest head’s recovery median is within the econometric reference range, while the under-dispersed head is slower and more often censored. ES comparisons are limited by the amount of tail information. With a median effective count near 26, the evidence supports broad groupings more than fine loss rankings.

In practice, the output form is checked first, then the extraction rule, then validation; repair is the last step. A head that cannot express the requested tail should not enter a native tail-risk workflow. A head that can express it must still pass backtests conditioned on the extraction rule. If a scale substitution is needed, the product should be described as a risk model whose location comes from the TSFM and whose scale comes from GARCH.

Data and code availability

The replication repository is public at <https://github.com/jianxhou/tsfm-tail-risk>. It contains the data fetchers and manifests, pinned model revisions, the extraction rules, the backtest battery with its vendored reference fixtures, all run scripts, and the script-generated results behind every table and figure.

Declaration of competing interest

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Appendix A. Run identity and pinned environment

The full grid is identified by configuration hash `d8d9e7c8f58dc5d2` with global seed `20260706`; model revisions are the 40-character commits abbreviated in Table 1. The analysis protocol fixed the extraction rules, Monte Carlo calibration, synthetic arm, reporting conventions, and contamination checks before the corresponding analyses were run. The results above were produced under that protocol; the replication package contains the run manifests, data manifests with per-series SHA-256 snapshots, and lock files.

Appendix B. Statistical validation against external references

The test suite contains 139 collected tests; its core statistical routines are checked against external fixtures. Kupiec and Christoffersen are checked against `rugarch` and GAS-compatible references; ESR and the exceedance tests are checked against `esback`, while FZ0 is checked against the QuantLet `HMD_ES` replication code, commit `2490253e`. The documented `kupiec.p` divergence is discussed in the footnote to Section 5; these checks use the pinned reference stack R 4.4.2, `rugarch` 1.5.5, and `esback` 0.3.1. The DQ implementation is validated by formula-level and seeded size-property tests, and no external DQ fixture is claimed. Because GARCH-EVT is the comparator that the headline repair is measured against, it is also checked against an external

McNeil–Frey fixture built from `evir` 1.7.4 and `rugarch`. The constrained CAViaR-SAV estimator of Section 5 is screened per window for path validity; on the evaluation panel it records zero failed fits and zero invalid forecast days, and the per-asset diagnostics are included in the replication package. On the reference input, our POT–GPD tail reproduces `evir`’s $\hat{\xi} = 0.2317$ and its 5% POT quantile -1.6863 (residual scale) exactly. Our GARCH- t filter matches `rugarch`’s one-step $\sigma = 0.4892$ with a measured relative difference of 0.39%, and the fixture has 5 dedicated tests. Python-side computation uses `statsmodels` (Seabold and Perktold, 2010) and `arch` (Sheppard, 2025); R-side references are Galanos (2026); Bayer and Dimitriadis (2023); Pfaff and McNeil (2018).

The rest of this section gives the reconciliation detail for the unresolved within-family discrepancy with the conformal-shift recalibration study discussed in Section 2. For their quantile-grid heads, the deep tails are extracted from a nine-decile grid under a parametric tail completion, varied only as a robustness check. The closure analysis reports that “All closures yield $R > 1$ for both models” and cautions that “the replacement-regime classification therefore depends on the extrapolation rule, and a single rule should not be load-bearing.” Our probes instead request deep quantiles from the deployed Moirai-2.0 interface directly and obtain distinct monotone values at 1%/2.5%/5% (Section 3).

They document raw 1% violation rates of “approximately 99%” for TimesFM-2.5 and Moirai-2.0 and defend them as genuine. On our 32-asset evaluation panel, Moirai-2.0’s native deep tail violates at 1.39%. TimesFM-2.5’s extraction-based violation rates are $1.04\text{--}3.40\times$ nominal. The TimesFM-2.5 divergence is the larger of the two: that head exposes no native deep tail, so every deep-tail number in either study is the product of a completion or extraction layer. As a check, we apply their nine-decile Student- t completion protocol to our stored TimesFM-2.5 forecast grids on all 32 assets. The resulting 1% violation rates have mean 1.88% and range 0.91–2.69%, far below the reported magnitude on every asset. The head-type annotations also differ: their model table lists Moirai-2.0 as a mixture head with a quantile-grid output format, whereas our pinned-revision probe characterizes it as a quantile head.

Appendix C. Synthetic-DGP arbitration arm

Twenty GARCH- t and twenty GJR- t paths calibrated to SPX, sixty sampled windows each, have conditional tails known in closed form. The arm covers the four quantile heads (Chronos-Bolt, Chronos-2, TimesFM-2.5, Moirai-2.0) and arbitrates between E2 and E3 only; Lag-Llama’s parametric head and the rules E0 and E1 are not covered. Each model forecasts zero-shot on these paths and each extraction rule’s output is scored against the analytic truth. At $\alpha = 1\%$ the GPD splice (E3) is the more faithful VaR extraction rule for each of the four audited heads; the ES advantage holds across the four-head E2-versus-E3 grid and is larger than the VaR advantage (mean absolute ES errors roughly 1.2–1.4 return points under E3 versus 1.6–2.1 under E2, across those heads). Moirai-2.0 is the example cell of that comparison (its $\alpha = 1\%$ arbitration selects E3). At $\alpha \in \{2.5\%, 5\%\}$ the t -tail E2 is mildly closer for the two Chronos heads on VaR, which is one reason all headline results are conditioned on the extraction rule and not restricted to E3. Full per-head, per- α fidelity tables are script-generated in the replication package; the arbitration is valid under the synthetic DGP only (Section 12).

Appendix D. Full tables

Table D.1 lists the 32-asset evaluation panel. Table D.2 reports the complete repair matrix, with both coverage and FZ0 for E3 and E2 on all five models. Table D.8 gives the strict-ESR grid over E1–E3 and the econometric models, Table D.9 the Acerbi–Székely Z_2 summary, and Table D.3 the F2 step-size sweep. Table D.4 contains the Model Confidence Set membership rates at the two registered levels, and Table D.5 the corresponding repair-battery memberships. Table D.6 reports the conditional-coverage and dynamic-quantile pass rates that accompany the unconditional-coverage audit of Section 6. Table D.10 decomposes the pooled 96-cell audit rate by tail level at E3 and adds the fraction of assets not rejected at all three levels simultaneously. Table D.11 reports the late-start matched-date sensitivity, and Table D.12 the pretraining-bound exposure shares. Table D.13 reports the leave-one-class-out and unclustered-bootstrap robustness of the decomposition intervals, and Table D.15 the Newey–West sensitivity of the Diebold–Mariano comparisons. Table D.16 reports the threshold-sensitivity grid for the joint criterion: H passes on all five models in 14 of 36 combinations; no model passes at FZ0 tolerance 1.00.

Table D.1: The 32-asset evaluation panel: series key, asset class, raw sample span, return transform (log-return $\times 100$ for prices; daily basis-point first differences for Treasury yields), and public source. Script-generated from the data manifest; the two quarantined FX pairs and the two regime-dating volatility indices are excluded (Section 4).

key	class	start	end	transform	source
csi300	equity index	2013-11-06	2026-07-02	logret_x100	yahoo
dax	equity index	2000-01-03	2026-07-03	logret_x100	yahoo
ftse	equity index	2000-01-04	2026-07-03	logret_x100	yahoo
hsi	equity index	2000-01-03	2026-07-03	logret_x100	yahoo
ndx	equity index	2000-01-03	2026-07-02	logret_x100	yahoo
nikkei	equity index	2000-01-04	2026-07-03	logret_x100	yahoo
rut	equity index	2000-01-03	2026-07-02	logret_x100	yahoo
spx	equity index	2000-01-03	2026-07-02	logret_x100	yahoo
aapl	single stock	2000-01-03	2026-07-02	logret_x100	yahoo
amzn	single stock	2000-01-03	2026-07-02	logret_x100	yahoo
jnj	single stock	2000-01-03	2026-07-02	logret_x100	yahoo
jpm	single stock	2000-01-03	2026-07-02	logret_x100	yahoo
msft	single stock	2000-01-03	2026-07-02	logret_x100	yahoo
nvda	single stock	2000-01-03	2026-07-02	logret_x100	yahoo
wmt	single stock	2000-01-03	2026-07-02	logret_x100	yahoo
xom	single stock	2000-01-03	2026-07-02	logret_x100	yahoo
audusd	FX	2006-05-16	2026-07-04	logret_x100	yahoo
gbpusd	FX	2003-12-01	2026-07-04	logret_x100	yahoo
usdcad	FX	2003-09-17	2026-07-04	logret_x100	yahoo
usdchf	FX	2003-09-17	2026-07-03	logret_x100	yahoo
btc	crypto	2014-09-17	2026-07-04	logret_x100	yahoo
eth	crypto	2017-11-09	2026-07-04	logret_x100	yahoo
ltc	crypto	2014-09-17	2026-07-04	logret_x100	yahoo
xrp	crypto	2017-11-09	2026-07-04	logret_x100	yahoo
brent	commodity	2007-07-30	2026-07-03	logret_x100	yahoo
copper	commodity	2000-08-30	2026-07-03	logret_x100	yahoo
gold	commodity	2000-08-30	2026-07-03	logret_x100	yahoo
natgas	commodity	2000-08-30	2026-07-03	logret_x100	yahoo
dgs10	rates	2000-01-03	2026-07-02	diff_bp	treasury
dgs2	rates	2000-01-03	2026-07-02	diff_bp	treasury
dgs30	rates	2000-01-03	2026-07-02	diff_bp	treasury
dgs5	rates	2000-01-03	2026-07-02	diff_bp	treasury

Table D.2: Repair matrix for all five heads at $\alpha = 1\%$, matched sample: coverage (fraction of assets passing MC UC+CC at 5%), mean FZ0 with its asset count n in parentheses, and the paired difference $\Delta = \text{mean}(\text{FZ0}_{\text{arm}} - \text{FZ0}_{\text{unrepaired}})$ over the assets where both are finite (common n in parentheses; negative favors the arm), E3 and E2 extraction side by side. Cross-arm level comparisons should read Δ , not the bare means: the asset sets differ by arm in three distinct ways. Unrepaired, F2, and F3 lose dgs2 and dgs30 ($n=30$), where $\xi \geq 1$ windows leave the E3 ES undefined on more than 1% of matched days, failing the 99% ES-validity gate; F1 and H replace the extraction tail, so their ES is valid on all 32 assets (and no E2 column applies); F4’s additive shift breaks $e < v$ on more than 1% of days in every excluded cell, and its retained assets — listed below the table — are endogenous to that failure mode. No displayed cell is flagged by the low-information screen at the matched n (median 2139).

head	arm	cov. E3	FZ0 E3 (n)	Δ vs unrepaired. (n)	cov. E2	FZ0 E2 (n)
Chronos-2	unrepaired	69%	1.837 (30)	—	53%	1.907 (32)
Chronos-2	F1	78%	1.877 (32)	−0.022 (30)	—	—
Chronos-2	F2	72%	1.836 (30)	−0.001 (30)	56%	1.904 (32)
Chronos-2	F3	53%	1.887 (30)	+0.050 (30)	41%	1.948 (32)
Chronos-2	F4	78%	2.166 (10)	−0.007 (10)	72%	2.188 (15)
Chronos-2	H	69%	1.850 (32)	−0.052 (30)	—	—
Moirai-2.0	unrepaired	59%	1.836 (30)	—	53%	1.918 (32)
Moirai-2.0	F1	62%	1.899 (32)	+0.002 (30)	—	—
Moirai-2.0	F2	62%	1.837 (30)	+0.000 (30)	50%	1.916 (32)
Moirai-2.0	F3	50%	1.926 (30)	+0.089 (30)	44%	1.995 (32)
Moirai-2.0	F4	53%	2.010 (9)	+0.069 (9)	56%	2.454 (7)
Moirai-2.0	H	72%	1.851 (32)	−0.051 (30)	—	—
Lag-Llama	unrepaired	6%	2.148 (30)	—	16%	2.170 (32)
Lag-Llama	F1	69%	2.008 (32)	−0.208 (30)	—	—
Lag-Llama	F2	6%	2.056 (30)	−0.092 (30)	22%	2.106 (32)
Lag-Llama	F3	31%	2.040 (30)	−0.108 (30)	22%	2.134 (32)
Lag-Llama	F4	50%	1.992 (12)	−0.131 (12)	44%	2.062 (9)
Lag-Llama	H	81%	1.856 (32)	−0.357 (30)	—	—
Chronos-Bolt	unrepaired	28%	1.897 (30)	—	56%	1.936 (32)
Chronos-Bolt	F1	69%	1.884 (32)	−0.076 (30)	—	—
Chronos-Bolt	F2	38%	1.876 (30)	−0.021 (30)	56%	1.928 (32)
Chronos-Bolt	F3	50%	1.915 (30)	+0.018 (30)	34%	1.984 (32)
Chronos-Bolt	F4	59%	2.072 (10)	+0.018 (10)	59%	2.247 (15)
Chronos-Bolt	H	72%	1.849 (32)	−0.112 (30)	—	—
TimesFM-2.5	unrepaired	59%	1.846 (30)	—	72%	1.916 (32)
TimesFM-2.5	F1	59%	1.902 (32)	−0.006 (30)	—	—
TimesFM-2.5	F2	69%	1.843 (30)	−0.002 (30)	72%	1.915 (32)
TimesFM-2.5	F3	50%	1.920 (30)	+0.074 (30)	41%	1.992 (32)
TimesFM-2.5	F4	53%	1.906 (9)	+0.074 (9)	56%	2.027 (8)
TimesFM-2.5	H	72%	1.850 (32)	−0.062 (30)	—	—

F4 retained assets (E3, $\alpha = 1\%$): Chronos-2 (10): btc, dax, dgs10, dgs5, eth, ftse, hsi, ndx, wmt, xrp; Moirai-2.0 (9): csi300, dgs10, dgs5, ftse, hsi, ltc, ndx, usdchf, xom; Lag-Llama (12): audusd, csi300, dgs10, dgs5, ftse, hsi, ltc, ndx, spx, usdchf, wmt, xom; Chronos-Bolt (10): csi300, dax, dgs10, dgs5, eth, ftse, hsi, ndx, wmt, xom; TimesFM-2.5 (9): amzn, csi300, dgs10, dgs5, ftse, hsi, ndx, usdcad, wmt. All other assets are excluded by the $e < v$ failure above; the complement of each list is the excluded set.

Table D.3: F2 (adaptive conformal) step-size robustness at $\alpha = 1\%$, E3, matched sample: coverage and mean FZ0 under $\gamma \in \{0.5\times, 1\times, 2\times\}$ of the pre-specified scale $0.01\hat{\sigma}$.

head	cov. $0.5\times$	cov. $1\times$	cov. $2\times$	FZ0 $0.5\times$	FZ0 $1\times$	FZ0 $2\times$
Chronos-2	75%	72%	78%	1.836 (n=30)	1.836 (n=30)	1.835 (n=30)
Moirai-2.0	59%	62%	69%	1.836 (n=30)	1.837 (n=30)	1.837 (n=30)
Lag-Llama	6%	6%	16%	2.092 (n=30)	2.056 (n=30)	2.010 (n=30)
Chronos-Bolt	31%	38%	47%	1.885 (n=30)	1.876 (n=30)	1.864 (n=30)
TimesFM-2.5	66%	69%	69%	1.844 (n=30)	1.843 (n=30)	1.842 (n=30)

Table D.4: Model Confidence Set membership rate (fraction of assets on which the forecaster is in the MCS by FZ0 loss) at the two registered levels, 10% and 25%, E2 extraction for TSFMs, by α . Higher = more often statistically indistinguishable from the best; the 25% level eliminates more aggressively, so its rates bound the 10% rates from below. ‘—’: CAViaR-SAV forecasts VaR only, so FZ0 loss is undefined.

forecaster	10% level			25% level		
	$\alpha = 1\%$	2.5%	5%	$\alpha = 1\%$	2.5%	5%
Chronos-Bolt	100%	100%	91%	94%	88%	75%
Chronos-2	100%	100%	97%	97%	81%	66%
TimesFM-2.5	97%	97%	91%	97%	88%	75%
Moirai-2.0	97%	97%	88%	91%	81%	62%
Lag-Llama	50%	22%	9%	34%	6%	6%
GARCH(1,1)- t	100%	100%	100%	97%	97%	94%
GJR-GARCH- t	100%	100%	100%	97%	100%	100%
EWMA(0.94)	100%	100%	100%	75%	94%	94%
HS-250	84%	75%	69%	62%	50%	28%
HS-500	75%	62%	44%	50%	31%	19%
FHS	100%	100%	100%	100%	97%	94%
CAViaR-SAV	—	—	—	—	—	—

Table D.5: Repair-battery Model Confidence Sets at the two registered levels: share of the 160 model–asset cells (5 heads $\times$ 32 assets) in which each arm survives the FZ0-loss MCS over the seven-member set {F1, F2, F3, F4, H, GARCH–EVT, FHS}, matched sample, by α .

arm	10% level			25% level		
	$\alpha = 1\%$	2.5%	5%	$\alpha = 1\%$	2.5%	5%
F1	81%	78%	69%	66%	60%	50%
F2	79%	76%	76%	57%	58%	58%
F3	50%	52%	42%	35%	34%	28%
F4	64%	60%	59%	41%	41%	33%
H	99%	98%	92%	97%	92%	82%
GARCH–EVT	100%	99%	98%	97%	95%	94%
FHS	99%	99%	99%	91%	92%	90%

Table D.6: Conditional-coverage (CC, Christoffersen; MC-calibrated) and dynamic-quantile (DQ, four lags; asymptotic p) pass rates on the main 32-asset panel, pooled over asset $\times$ α cells per head and extraction rule, at the 5% test level. ‘param’ = econometric baselines’ native parametric forecasts (structure as in Table 2); TimesFM-2.5’s E0 dash follows the same native-head restriction noted there. The DQ columns rest on the asymptotic χ^2 reference, which over-rejects at deep tails: under the iid null at $n = 2,639$, $\alpha = 1\%$, the 5%-level asymptotic DQ test rejects 11.8% of the time, so DQ levels are comparable across forecasters but not sized at 5%. The companion unconditional-coverage table is Table 2.

forecaster	CC pass (MC)					DQ pass (asym.)				
	E0	E1	E2	E3	param	E0	E1	E2	E3	param
Chronos-Bolt	0%	2%	34%	16%	—	0%	1%	16%	8%	—
Chronos-2	66%	2%	50%	76%	—	40%	1%	25%	43%	—
TimesFM-2.5	—	5%	57%	54%	—	—	2%	26%	27%	—
Moirai-2.0	45%	21%	47%	74%	—	22%	10%	23%	41%	—
Lag-Llama	0%	0%	5%	0%	—	0%	0%	2%	1%	—
GARCH(1,1)- t	—	—	—	—	59%	—	—	—	—	50%
GJR-GARCH- t	—	—	—	—	68%	—	—	—	—	58%
EWMA(0.94)	—	—	—	—	43%	—	—	—	—	25%
HS-250	—	—	—	—	22%	—	—	—	—	3%
HS-500	—	—	—	—	29%	—	—	—	—	2%
FHS	—	—	—	—	91%	—	—	—	—	53%
CAViaR-SAV	—	—	—	—	86%	—	—	—	—	46%

Table D.7: A controlled forecast-form contrast inside the econometric battery: EWMA(0.94) and FHS share the same conditional-scale recursion ($\lambda = 0.94$, seeded at the window sample variance; the identity is pinned by a unit test), and differ only in tail form, a Gaussian quantile versus the filtered empirical tail. At $\alpha = 1\%$ the tail form alone moves the MC-UC pass count from 2/32 to 32/32; mean FZ0 and strict-ESR non-rejection move with it. All three columns are computed at $\alpha = 1\%$ over the 32 assets. Mean FZ0 is shown at four decimals because differences within the FZ0 loss cluster are resolved at that precision.

forecaster	UC pass ($\alpha = 1\%$)	mean FZ0 ($\alpha = 1\%$)	ESR non-rej. ($\alpha = 1\%$)
EWMA(0.94), normal tail	2/32	1.9604	12.5%
FHS, filtered empirical tail	32/32	1.8140	81.2%

Table D.8: Strict ESR backtest: fraction of the 96 asset–tail cells (32 assets $\times$ three α levels) at which the two-sided asymptotic ESR test does not reject at the 5% level, by forecaster and extraction rule; E1 is the deliberately misspecified Gaussian control. ‘param’ = native econometric forecasts. CAViaR-SAV forecasts VaR only and has no ES, hence no ESR cell; E0 has no ES companion, hence no E0 column. Each cell’s test runs on that cell’s own ES-valid day set (invalid days removed; cells enter with at least 100 valid days and no coverage gate), so ESR denominators differ from the FZ0/ Z_2 validity sets stated in their tables. The engine is the vendored **esback** strict ESR (version 1), two-sided asymptotic p with the bootstrap disabled ($B = 0$) and the point-estimation layer seeded per cell, so reported p -values are exactly reproducible; unlike the UC/CC layer, this test is not MC-calibrated at deep-tail sample sizes. Descriptive summaries of cell-level decisions.

forecaster	E1	E2	E3	param
Chronos-Bolt	0.0%	45.8%	38.5%	—
Chronos-2	0.0%	34.4%	41.7%	—
TimesFM-2.5	2.1%	80.2%	82.3%	—
Moirai-2.0	6.2%	49.0%	85.4%	—
Lag-Llama	0.0%	16.7%	11.5%	—
GARCH(1,1)- t	—	—	—	89.6%
GJR-GARCH- t	—	—	—	89.6%
EWMA(0.94)	—	—	—	8.3%
HS-250	—	—	—	28.1%
HS-500	—	—	—	43.8%
FHS	—	—	—	78.1%
CAViaR-SAV	—	—	—	—

Table D.9: Acerbi–Székely Z_2 statistic, summarized over the asset–tail cells per forecaster and extraction rule: cell count, mean and median Z_2 , and the share of cells with $Z_2 > 0$. $Z_2 = 0$ under a correct (VaR, ES) pair; $Z_2 > 0$ indicates realized tail losses exceeding the predicted ES. The statistic is reported descriptively, without a bootstrap p -value. E0 has no ES cell; E3 counts below 96 reflect the dgs30 cells whose GPD ES is genuinely undefined ($\xi \geq 1$); CAViaR-SAV has no ES output. The HS-250 count of 95 likewise reflects dgs30 ($\alpha = 1\%$): 1bp quantization ties the empirical ES to VaR on 70 of 2,625 days, below the 99% valid-ES-day guard.

forecaster	rule	cells	mean Z_2	median Z_2	share > 0
Chronos-Bolt	E1	96	2.525	1.921	100%
Chronos-Bolt	E2	96	0.301	0.303	91%
Chronos-Bolt	E3	93	0.530	0.507	95%
Chronos-2	E1	96	2.464	1.862	100%
Chronos-2	E2	96	0.206	0.212	86%
Chronos-2	E3	93	0.067	0.069	63%
TimesFM-2.5	E1	96	1.962	1.515	100%
TimesFM-2.5	E2	96	0.066	0.085	64%
TimesFM-2.5	E3	93	0.159	0.178	83%
Moirai-2.0	E1	96	1.311	0.995	98%
Moirai-2.0	E2	96	−0.235	−0.219	2%
Moirai-2.0	E3	93	−0.007	0.016	52%
Lag-Llama	E1	96	3.862	3.231	100%
Lag-Llama	E2	96	0.950	0.914	96%
Lag-Llama	E3	93	1.192	1.107	100%
GARCH(1,1)- t	param	96	0.232	0.211	86%
GJR-GARCH- t	param	96	0.238	0.225	88%
EWMA(0.94)	param	96	0.731	0.554	99%
HS-250	param	95	0.415	0.345	100%
HS-500	param	96	0.267	0.213	98%
FHS	param	96	0.047	0.029	76%

Table D.10: VaR audit by tail level, extraction rule E3: fraction of the 32 assets per α at which the Monte-Carlo-calibrated Kupiec unconditional-coverage test does not reject; the pooled 96-cell rate of Table 2; and the fraction of assets not rejected at all three levels simultaneously. Descriptive summaries of cell-level decisions, not jointly sized tests.

forecaster	pooled	$\alpha = 1\%$	$\alpha = 2.5\%$	$\alpha = 5\%$	all three
Chronos-Bolt	19.8%	34.4%	18.8%	6.2%	6.2%
Chronos-2	84.4%	87.5%	78.1%	87.5%	75.0%
TimesFM-2.5	69.8%	84.4%	68.8%	56.2%	40.6%
Moirai-2.0	91.7%	90.6%	87.5%	96.9%	81.2%
Lag-Llama	0.0%	0.0%	0.0%	0.0%	0.0%

Table D.11: Matched-date sensitivity on the five late-start assets (csi300, eth, xrp, btc, ltc), where the 1,000-day baseline estimation window starts 459–488 trading days after the 512-day TSFM context: MC-UC pass counts per head and rule over the 15 asset–tail cells, on the delivered TSFM day sets and on the intersection with the baseline day sets (existing forecasts only). ‘flips’ counts cells whose UC decision changes; the flips are few and not in one direction.

head	rule	cells	pass (delivered)	pass (matched)	flips
Chronos-Bolt	E0	15	0	0	0
Chronos-Bolt	E1	15	0	0	0
Chronos-Bolt	E2	15	11	12	1
Chronos-Bolt	E3	15	3	2	1
Chronos-2	E0	15	10	11	1
Chronos-2	E1	15	0	0	0
Chronos-2	E2	15	8	8	0
Chronos-2	E3	15	15	15	0
TimesFM-2.5	E1	15	0	0	0
TimesFM-2.5	E2	15	13	15	2
TimesFM-2.5	E3	15	13	13	2
Moirai-2.0	E0	15	3	2	1
Moirai-2.0	E1	15	1	1	0
Moirai-2.0	E2	15	3	5	2
Moirai-2.0	E3	15	14	15	1
Lag-Llama	E0	15	0	0	0
Lag-Llama	E1	15	0	0	0
Lag-Llama	E2	15	5	5	0
Lag-Llama	E3	15	0	1	1

Table D.12: Pretraining-bound exposure of the evaluation sample: per head, the registered calendar bound, total evaluation asset-days over the 32 assets, and the share of those days on or before the bound. Bounds follow two rules disclosed in Section 4: Chronos-Bolt uses its pinned-revision date, while Chronos-2, Moirai-2.0, and Lag-Llama use report-based month-end bounds; TimesFM-2.5’s bound is vendor-disclosed. Under a unified revision-date rule, Chronos-2’s bound would move to 2026-06-05 and its post-bound sample would shrink from 5,662 to 641 asset-days.

head	bound	asset-days	share $\leq$ bound
Chronos-Bolt	2025-11-21	87,013	94.07%
Chronos-2	2025-10-31	87,013	93.49%
TimesFM-2.5	2023-11-30	87,013	74.73%
Moirai-2.0	2025-11-30	87,013	94.24%
Lag-Llama	2023-10-31	87,013	73.91%

Table D.13: Robustness of the location–scale decomposition intervals ($\alpha = 1\%$, E3; same frozen per-asset paired FZ0 series as Table D.14, whose clustered six-class bootstrap uses seed 20260717). LOO = range of the mean over the six deterministic leave-one-asset-class-out subsets; boot = asset-level iid bootstrap 95% intervals (B=2,000, seed 20260722). The clustered design of Table D.14 is the headline specification.

head	LOO scale	boot scale	LOO location	boot location
Chronos-2	[+0.011, +0.034]	[+0.007, +0.047]	[+0.000, +0.006]	[−0.003, +0.010]
Moirai-2.0	[+0.037, +0.059]	[+0.029, +0.069]	[+0.002, +0.006]	[−0.001, +0.010]
Lag-Llama	[+0.139, +0.160]	[+0.123, +0.181]	[+0.006, +0.013]	[+0.001, +0.017]
Chronos-Bolt	[+0.020, +0.040]	[+0.018, +0.051]	[+0.001, +0.004]	[−0.001, +0.008]
TimesFM-2.5	[+0.037, +0.064]	[+0.033, +0.073]	[−0.001, +0.006]	[−0.005, +0.011]

Table D.14: Decomposition uncertainty at $\alpha = 1\%$, E3, matched sample: per-asset paired FZ0 differences, scale term (F1–H) and location term (H–GE), with asset-class cluster block-bootstrap 95% confidence intervals (six classes resampled with replacement, $B = 2,000$). The count column reports assets on which the scale term exceeds the location term. Scope: the evaluated one-step decomposition.

head	scale>loc	scale (F1–H) [95% CI]	loc (H–GE) [95% CI]
Chronos-2	22/32	+0.027 [−0.002, +0.055]	+0.004 [−0.004, +0.013]
Moirai-2.0	24/32	+0.048 [+0.011, +0.075]	+0.004 [−0.001, +0.010]
Lag-Llama	30/32	+0.152 [+0.124, +0.179]	+0.009 [+0.001, +0.018]
Chronos-Bolt	22/32	+0.034 [+0.003, +0.058]	+0.003 [−0.003, +0.008]
TimesFM-2.5	21/32	+0.052 [+0.012, +0.083]	+0.003 [−0.007, +0.013]

Table D.15: Diebold–Mariano variance sensitivity: of the 160 asset-level comparisons per pair and tail level, counts significant at the 5% test level under the baseline variance and under a Newey–West (Bartlett) variance with automatic lag $\lfloor 4(n/100)^{2/9} \rfloor$, split by direction (favors the comparator / favors F1; the comparator is GARCH–EVT or F4); ‘moved’ counts decisions that change. Pooled levels = the 2.5% and 5% tail levels; the two pooled-level F1-favoring exceptions (F1 vs GARCH–EVT) are the same set under both variances; of the six pooled-level F4-favoring exceptions (F1 vs F4) under the baseline variance, five remain under Newey–West, a strict subset. The baseline DM statistics were reproduced exactly (max deviation 5×10^{-16}) before the variant was computed. The baseline column retains the pre-frozen HLN-corrected $t(n-1)$ convention; the Newey–West column reports uncorrected normal p -values.

pair	α	baseline sig.		NW sig.		moved
		comp.	F1	comp.	F1	
F1 vs GARCH–EVT	1%	42	0	37	0	5
F1 vs GARCH–EVT	2.5%	55	1	48	1	7
F1 vs GARCH–EVT	5%	67	1	58	1	9
F1 vs F4	1%	0	57	0	53	4
F1 vs F4	2.5%	3	45	2	38	10
F1 vs F4	5%	3	36	3	29	7

Table D.16: Sensitivity of H’s joint success to the registered thresholds: number of heads (of 5) on which H passes the joint criterion, over condition-(a) margin $\{5, 10, 15\}$ pp $\times$ FZ0 tolerance $\{1.00, 1.025, 1.05, 1.10\} \times$ condition-(b) fraction $\{1/2, 2/3, 3/4\}$ on the matched sample. The registered combination (10 pp, 1.05, 2/3; boxed) gives 5/5; all five heads pass in 14 of 36 combinations; no head passes at tolerance 1.00; at a 5 pp margin the condition-(a) threshold rises to 73.1% and four heads fall below it (Chronos-2 at 69%; Moirai-2.0, Chronos-Bolt, and TimesFM-2.5 at 72%), so one of five heads passes and the 5/5 headline is margin-sensitive.

margin	frac	tol	1.00	1.025	1.05	1.10
5 pp	1/2	0	1	1	1	1
5 pp	2/3	0	0	1	1	1
5 pp	3/4	0	0	1	1	1
10 pp	1/2	0	5	5	5	5
10 pp	2/3	0	3	5	5	5
10 pp	3/4	0	0	5	5	5
15 pp	1/2	0	5	5	5	5
15 pp	2/3	0	3	5	5	5
15 pp	3/4	0	0	5	5	5

Table D.17: Post-hoc information-set arm: the two GARCH variants and FHS re-estimated on 512-observation rolling windows, matching the TSFM context length, against the registered 1,000-observation setting. Same out-of-sample days, refit cadence, and 32-asset panel; UC pass is the share of assets passing MC-calibrated unconditional coverage at the 5% test level; FZ0 is the mean joint VaR–ES loss at $\alpha = 1\%$. This arm was added after the main results (not pre-specified); the registered setting remains 1,000.

window	UC pass, $\alpha = 1\%$		UC pass, $\alpha = 2.5\%$		UC pass, $\alpha = 5\%$		FZ0, $\alpha = 1\%$	
	512	1,000	512	1,000	512	1,000	512	1,000
GARCH(1,1)- t	62%	75%	59%	62%	50%	59%	1.832	1.800
GJR-GARCH- t	59%	72%	59%	72%	50%	59%	1.820	1.799
FHS	97%	100%	100%	100%	100%	100%	1.825	1.814

Table D.17 reports the post-hoc information-set arm: the two GARCH variants and FHS re-estimated on rolling windows matching the TSFM context length, alongside the registered longer-window setting.

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